

GENIFY

LECTURE 14

26-August-2025

TOPICS COVERED:

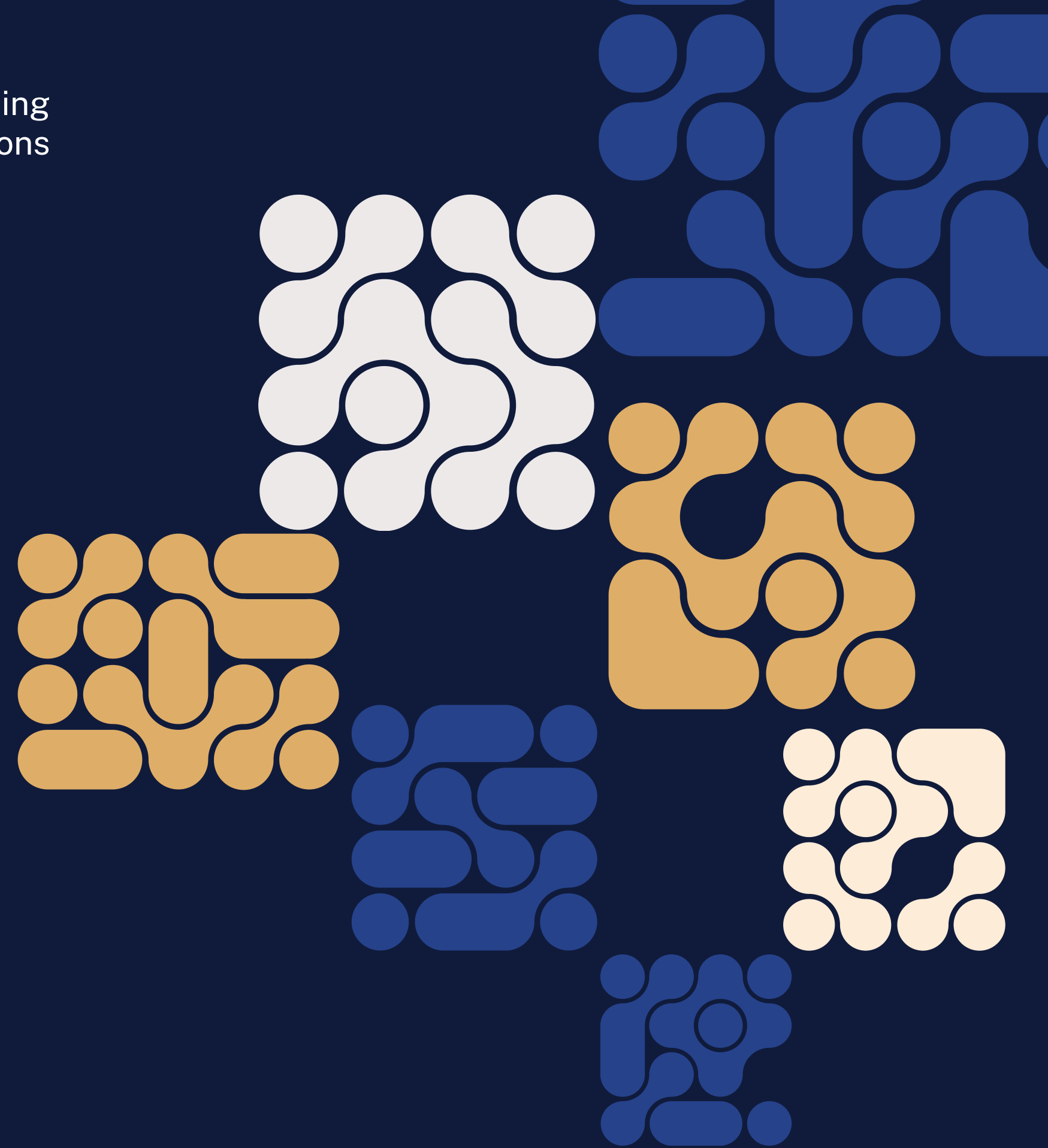
RIS

- TRANSFORMERS
 - TRANSFORMERS OVERVIEW
 - SELF ATTENTION

Presenter

Shahzaib Kashif

Co-Presenter

Shayan Hassan Baig

RECAP QUIZ TIME

ASSIGNMENT DISCUSSION



TRANSFORMERS

RNN Limitations

RNN Limitations

RNN Generated Output – Repitition

The cat is on the table. The cat is on the table. The cat sat on the table,

RNN Limitations

RNN Generated Output – Repetition

The cat is on the table. The cat is on the table. The cat sat on the table,

RNN Generated Output – Inappropriate

The boy went to the stores to buy some apples. He paid the cashier. The sun was shining brightly, and he ate the cake at the park.

RNN Limitations

RNN Generated Output – Repetition

The cat is on the table. The cat is on the table. The cat sat on the table,

RNN Generated Output – Inappropriate

The boy went to the stores to buy some apples. He paid the cashier. The sun was shining brightly, and he ate the cake at the park.

RNN Generated Output – Grammatically Incorrect

The group of people is walking in the park. They was excited about the picnic.

RNN Limitations

RNN Generated Poem

THE SUN IS SHINING BRIGHT,
IN THE SKY SO HIGH, IT TAKES FLIGHT.
THE BIRDS ARE SINGING, THE TREES STAND TALL,
BUT SOMETHING SEEMS OFF, NO SENSE AT ALL.

THE MOON IS RISING, THE STARS DO GLOW,
BUT DAY IS HERE, IT DOESN'T SHOW.
THE GRASS IS GREEN, THE WATER IS BLUE,
YET EVERYTHING FEELS OLD, AND SOMEHOW NEW.

RNN Generated Output – Repetition

The cat is on the table. The cat is on the table. The cat sat on the table,

RNN Generated Output – Inappropriate

The boy went to the stores to buy some apples. He paid the cashier. The sun was shining brightly, and he ate the cake at the park.

RNN Generated Output – Grammatically Incorrect

The group of people is walking in the park. They was excited about the picnic.

Generate a story of toys which are working hard in a toy-store factory

RNN Limitations

Alright, here's a short imaginative story for you:

The Midnight Factory

When the lights in *Maplewood Toy Store* flicked off and the last human footsteps faded into the night, the shelves came alive.

The stuffed bears stretched their arms, the wooden trains clicked their wheels, and the robots blinked awake with glowing eyes. Deep inside the store was a secret—an entire toy factory run by the toys themselves.

The factory wasn't for making *new* toys; it was for fixing and polishing the old ones. Broken wheels, scratched paint, or loose stitching—every flaw was taken to the **Midnight Workshop**.

At the front of the line stood **Captain Cog**, a wind-up tin soldier with a shiny red jacket. He barked orders like a factory foreman:

"Alright, crew! Teddy team—sewing station! Trains—off to the track repair zone! Robots—engines on, we need power for the assembly line!"

The teddy bears, with needles twice their size, carefully stitched each other's seams under the soft glow of fairy lights. The trains rolled along conveyor belts, carrying buckets of paint and varnish to freshen up dollhouses and rocking horses. The robots spun their gears to power the machines, huffing like steam engines.

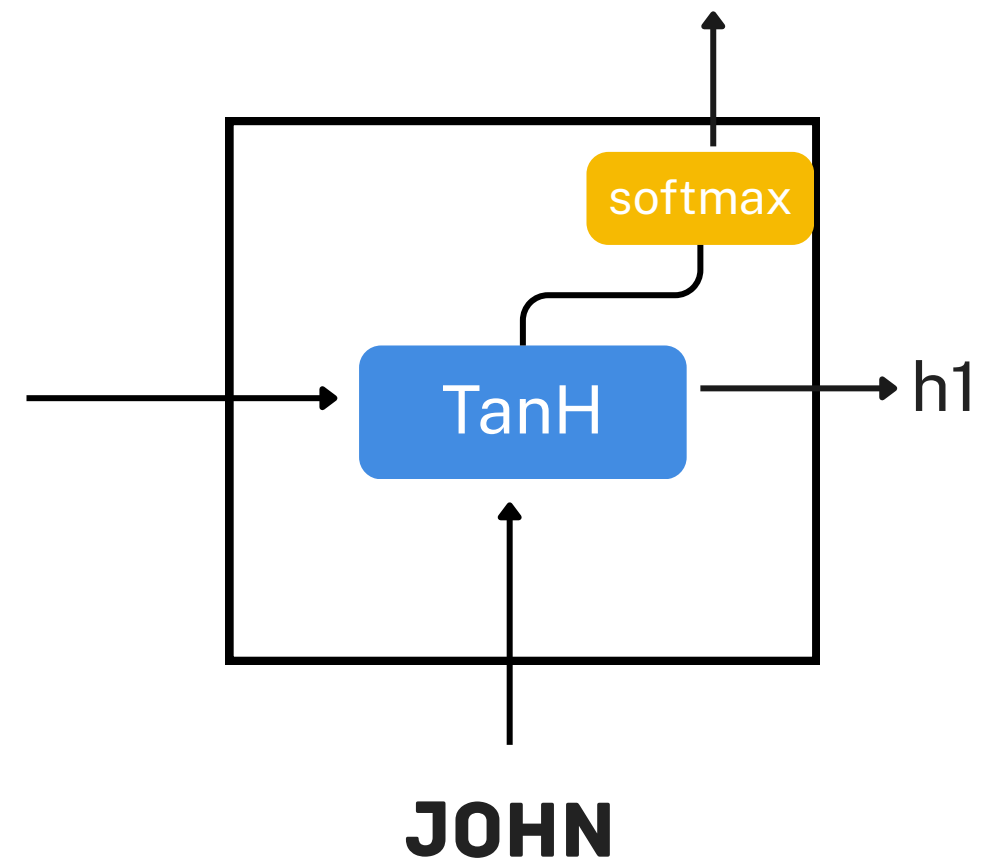
At the center of it all was **Luma**, a glass snow globe who kept time with her magical sparkle. Every midnight, she would hum a soft tune that set the rhythm of the work.

But tonight was special. The toys were preparing for the **Annual Children's Parade**—the big holiday

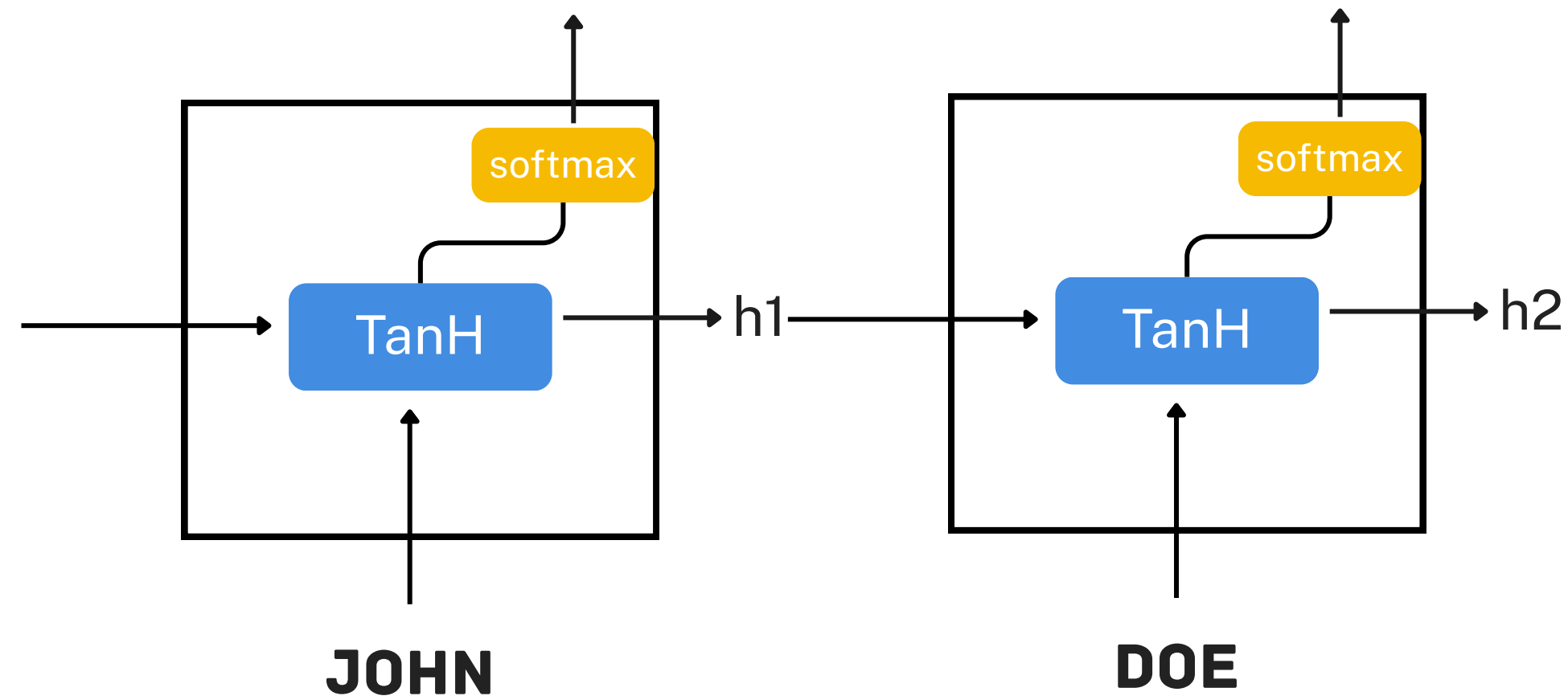
+ Ask anything



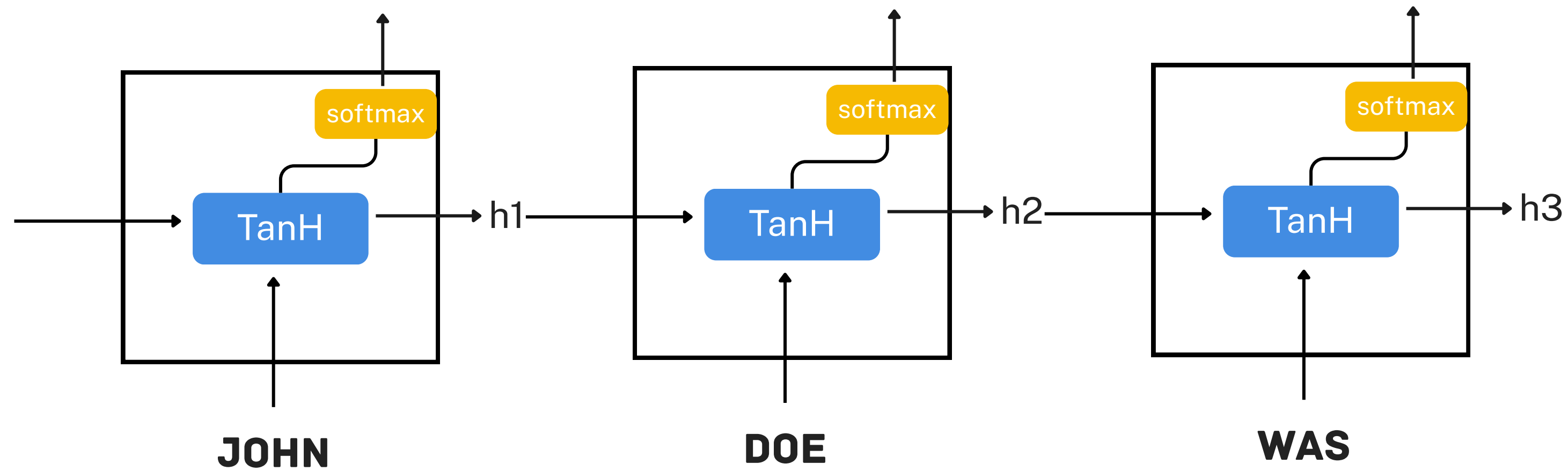
RNN Problems



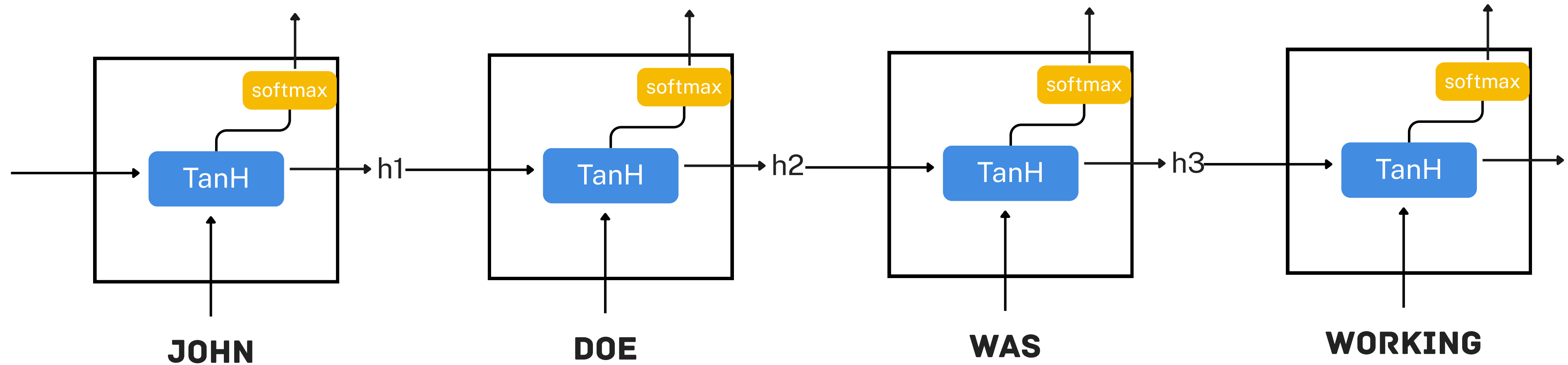
RNN Problems



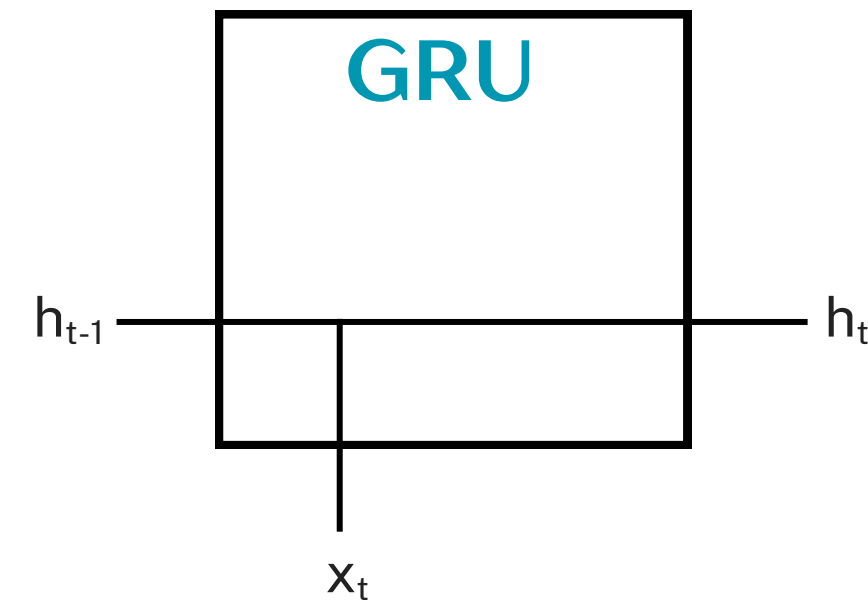
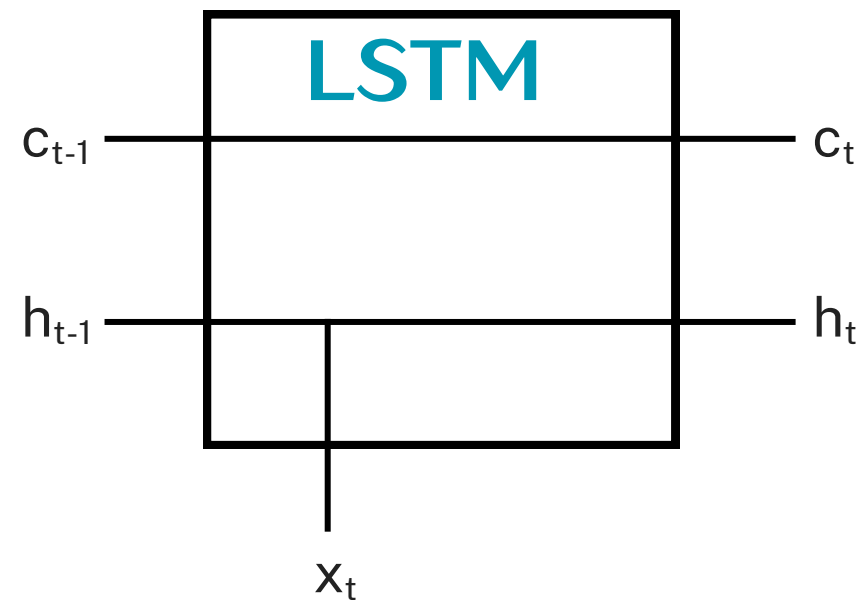
RNN Problems



RNN Problems



RNN Problems



When Jake first laid eyes on the Apple iPhone, he wasn't expecting much. It felt like holding the future. His old phone had served him well, and he wasn't one to chase the latest tech trends. But the moment he had the sleek, glassy device in his hand, there was something undeniably different about it.

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Before long, Jake was completely hooked. The iPhone wasn't just a phone anymore; it had become an extension of him. He loved how easily it synced with his computer, how effortlessly, he could check emails, and how FaceTime made talking to family feel more personal. It wasn't just the features, though. It was the experience. Every time he picked up the phone, it was like a little piece of technology, not just for what it could do, but for how it made his everyday life smoother, more exciting, and somehow more human.

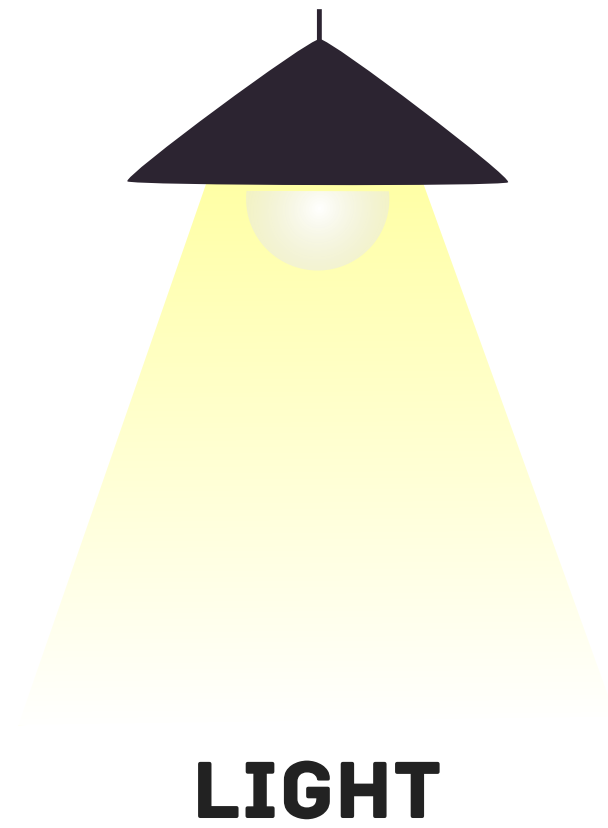
As he started exploring, he was amazed at how intuitive everything was. Jake marvelled at the seamless transitions between apps, the pinch to zoom function, and the way the phone seemed to understand what he wanted to do before he did. Then he discovered the camera. He had never been one for photography, but the clarity and ease of snapping pictures with the iPhone's camera had him capturing every moment around him. The built-in features, like visual voicemail and the music player, made him feel more connected and streamlined in ways he hadn't anticipated.

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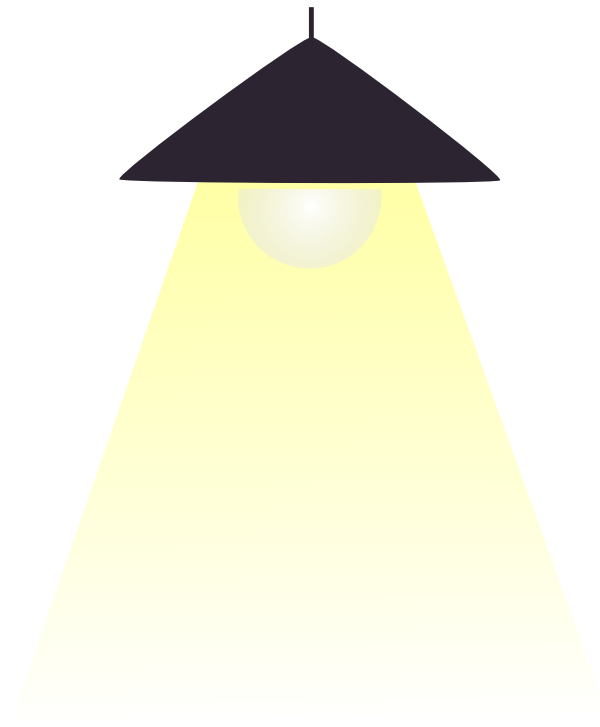
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RNN Problems



RNN Problems

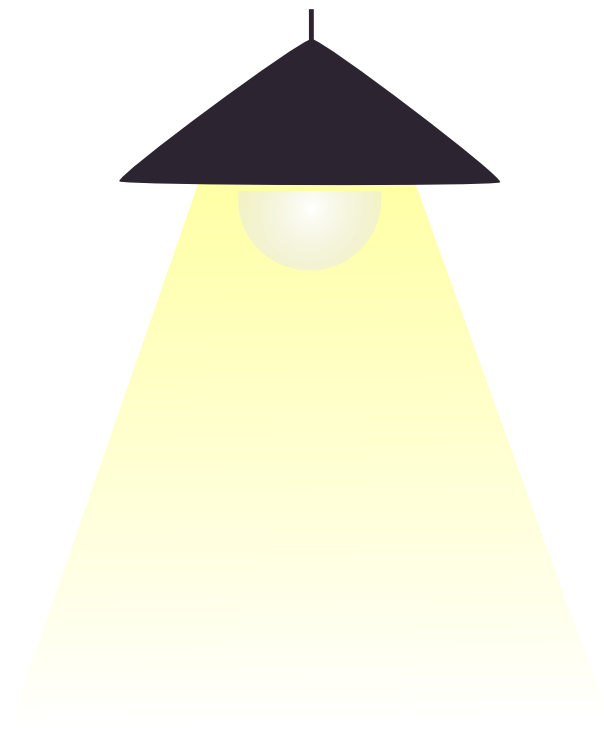


LIGHT



LIGHT WEIGHT

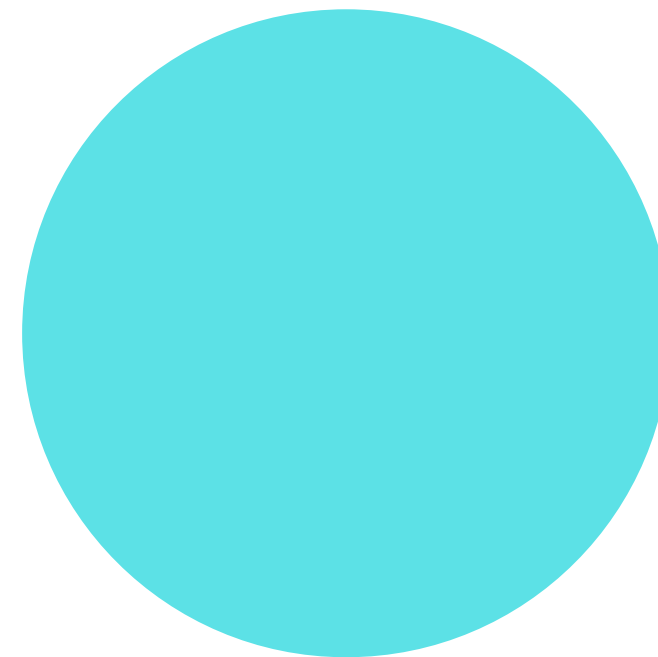
RNN Problems



LIGHT

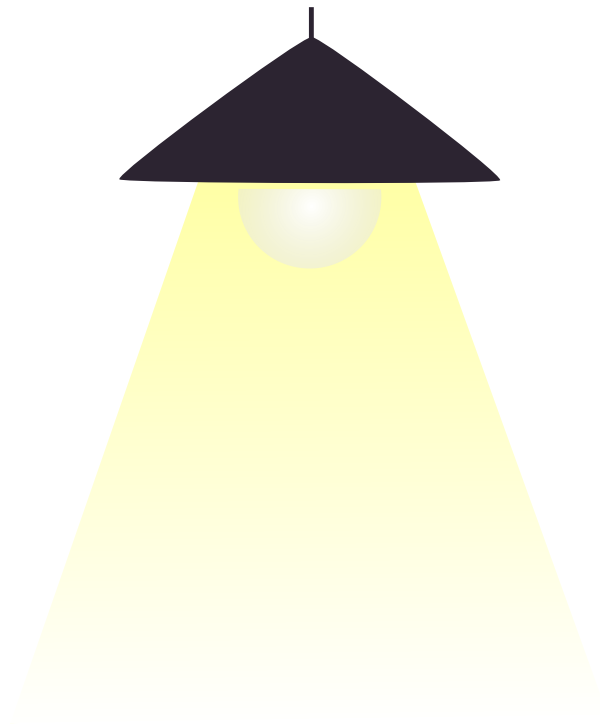


LIGHT WEIGHT



LIGHT BLUE

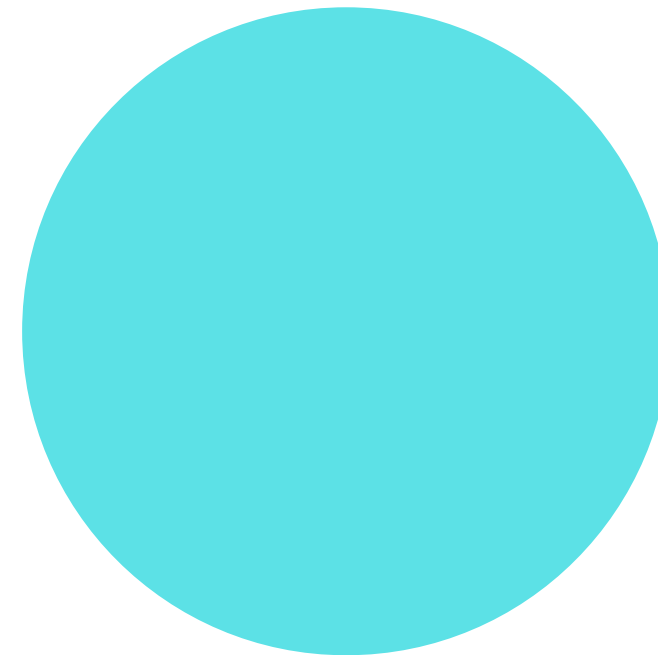
RNN Problems



LIGHT



LIGHT WEIGHT



LIGHT BLUE



LIGHT A CANDLE

RNN Problems - Word Embeddings

isTech	0.2
canCall	0.1
isSmart	0.19
.	.
.	.
isFruit	0.3
isFood	0.18

APPLE

RNN Problems - Word Embeddings

isTech	0.2
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APPLE

I LOVE APPLE PHONES



RNN Problems - Word Embeddings

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APPLE

I LOVE APPLE PHONES



I LOVE APPLE JUICE



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APPLE

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APPLE

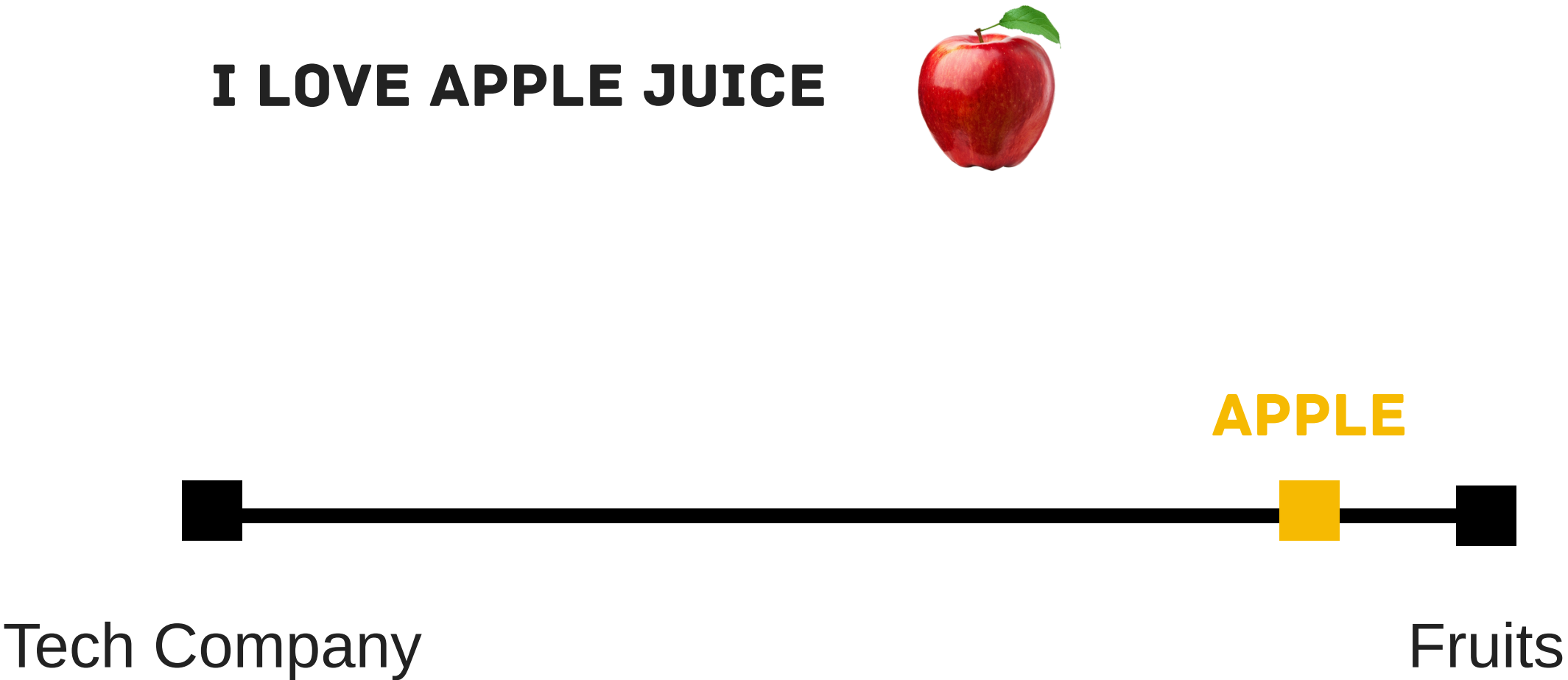
Tech Company

Fruits

RNN Problems - Word Embeddings

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APPLE

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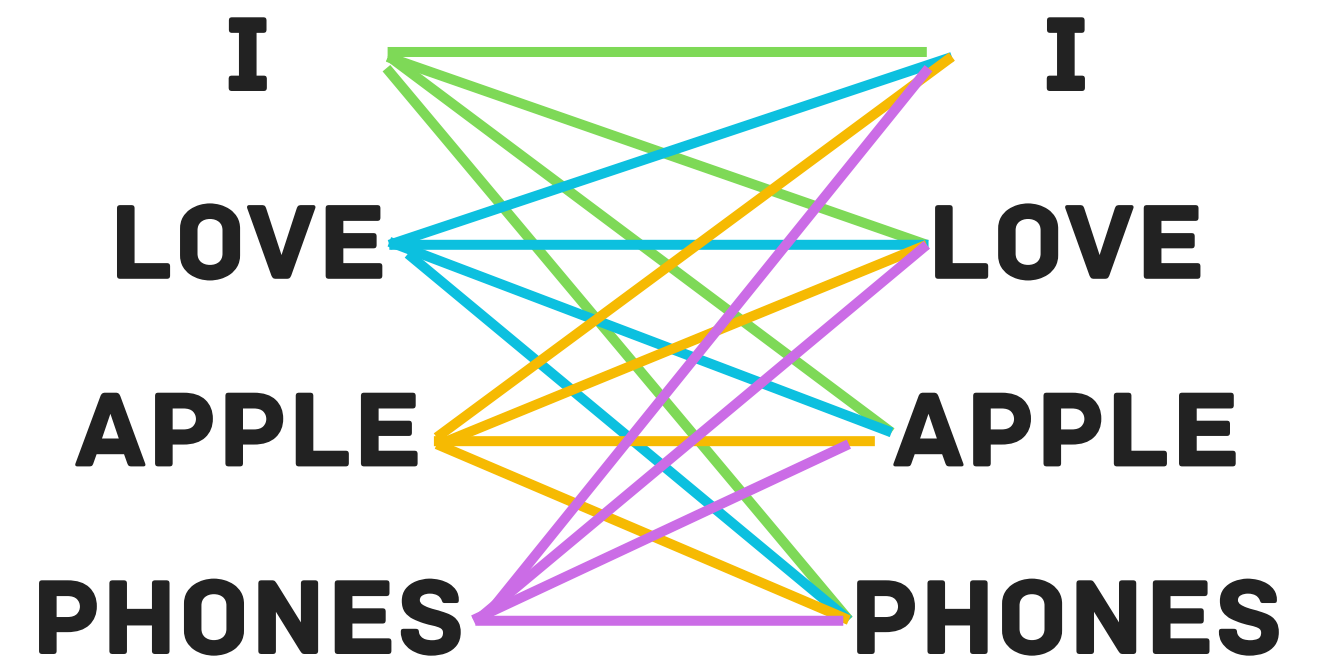
APPLE

Tech Company

Fruits

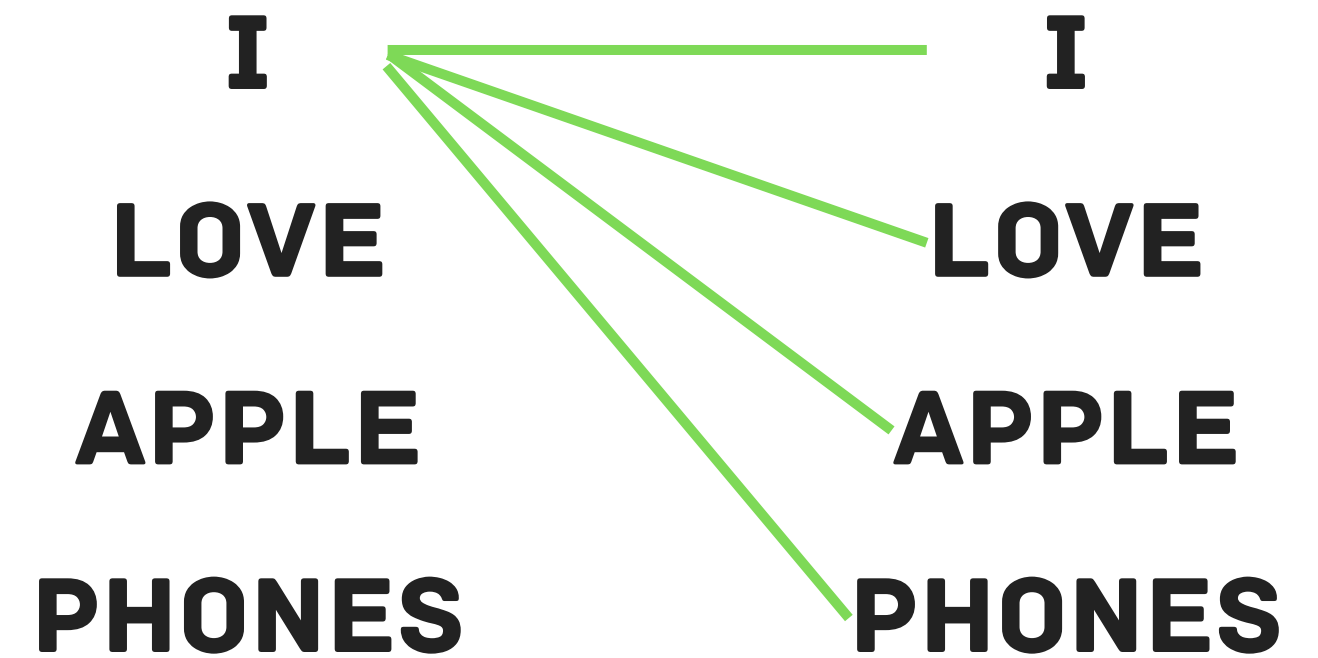
Self Attention - Brief Overview

SELF ATTENTION



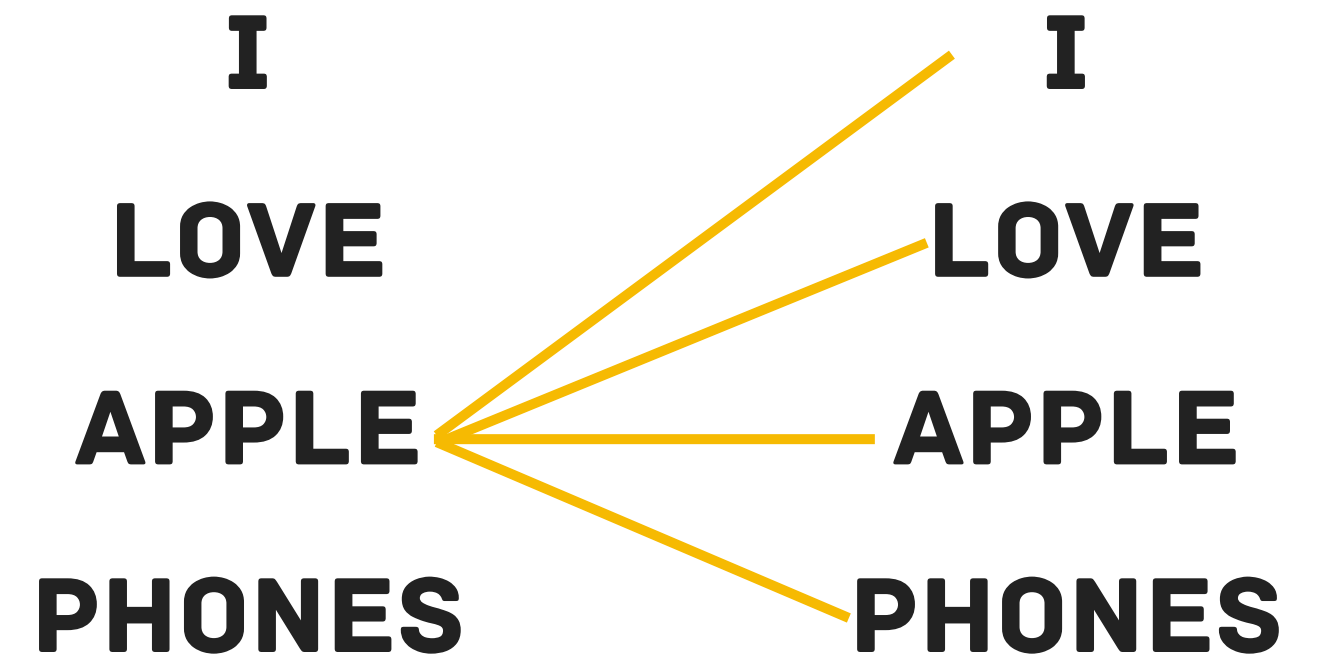
Self Attention - Brief Overview

SELF ATTENTION

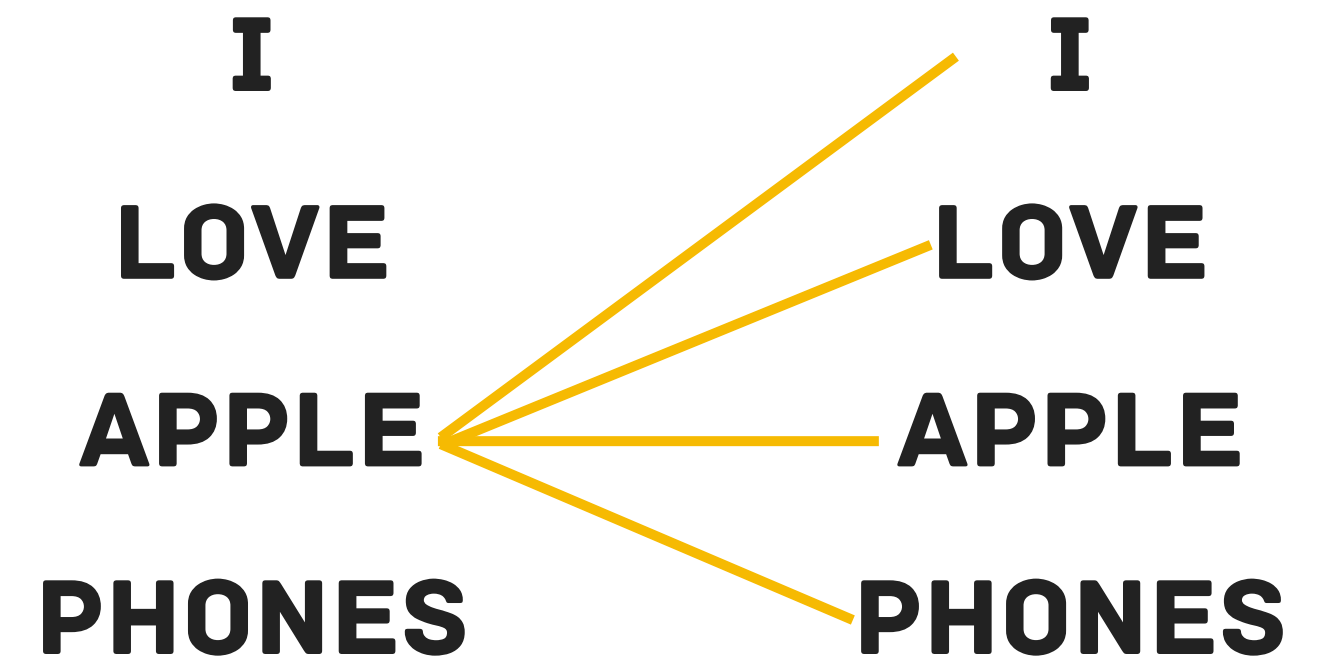
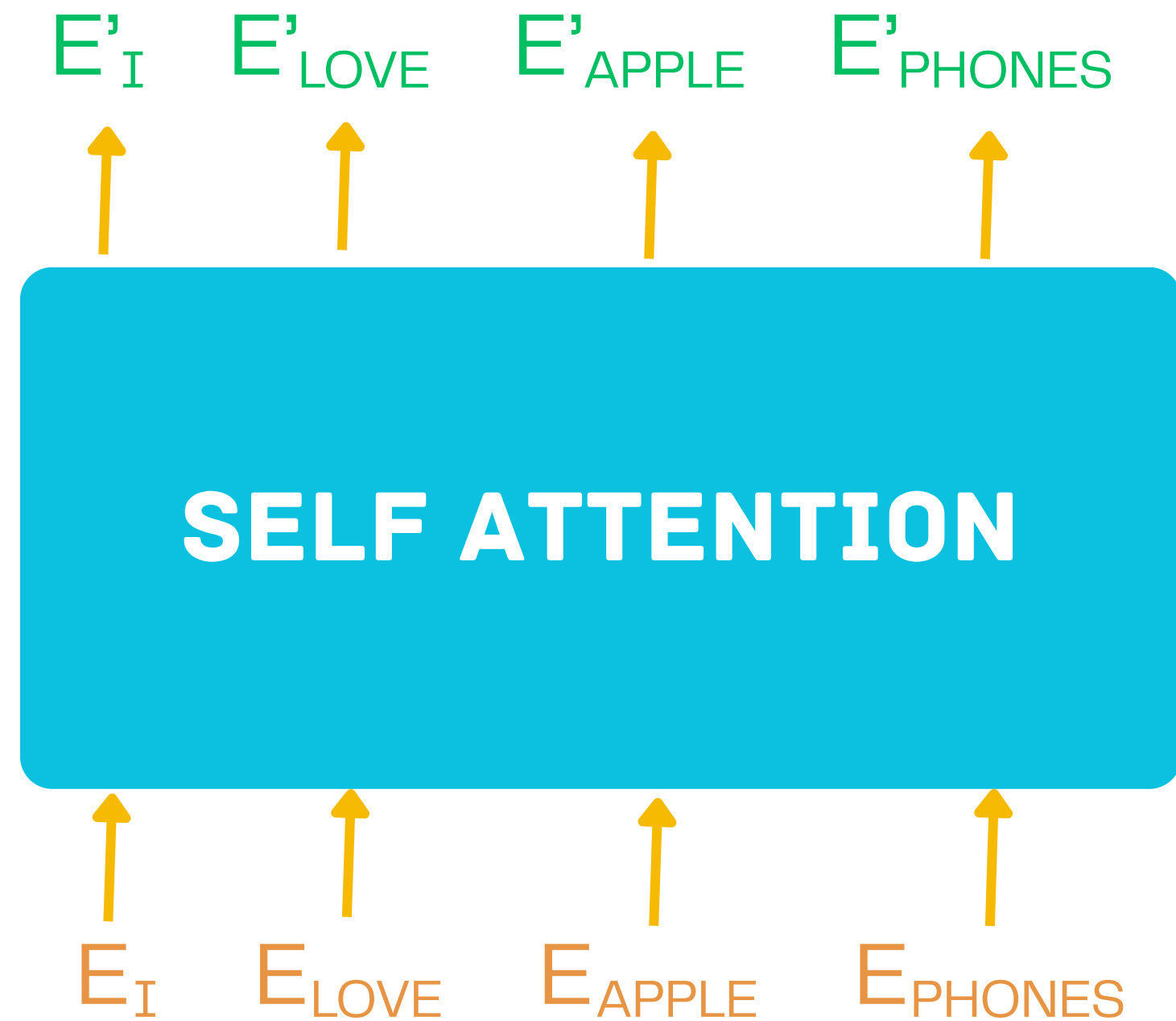


Self Attention - Brief Overview

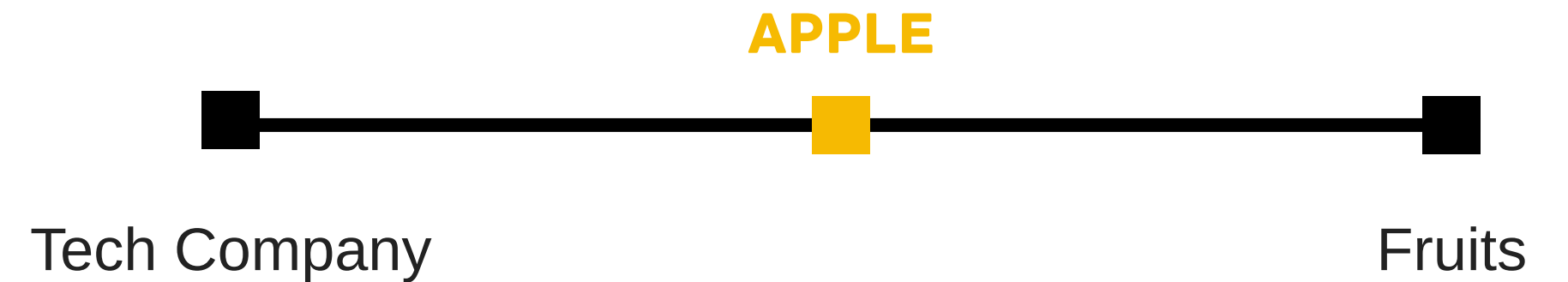
SELF ATTENTION



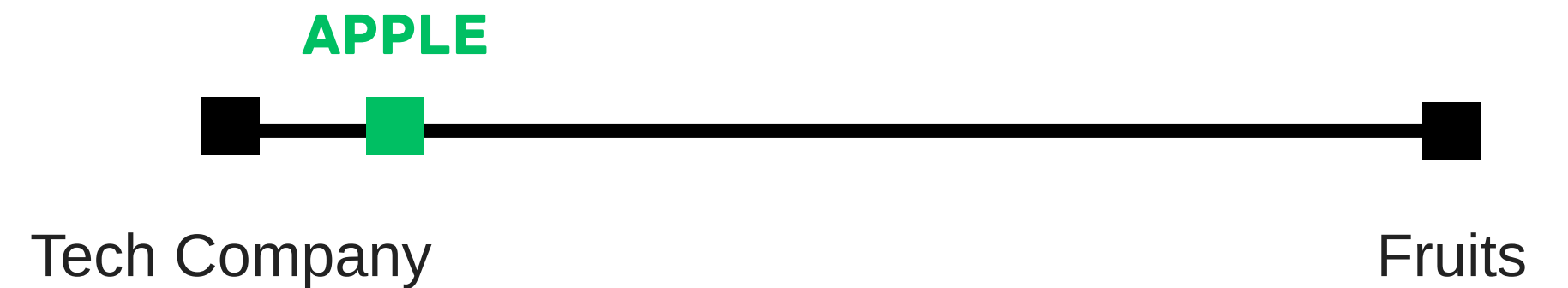
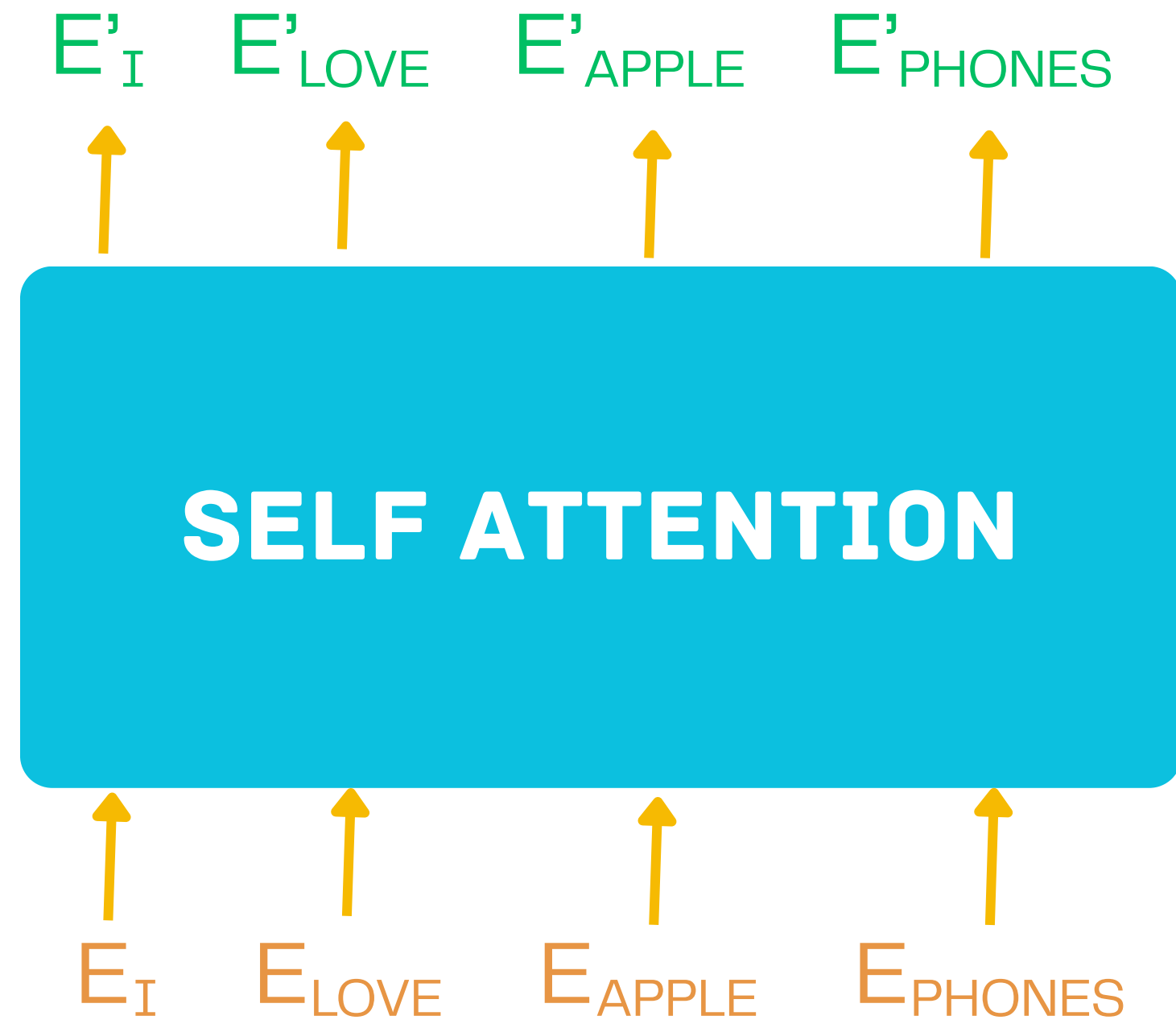
Self Attention - Brief Overview



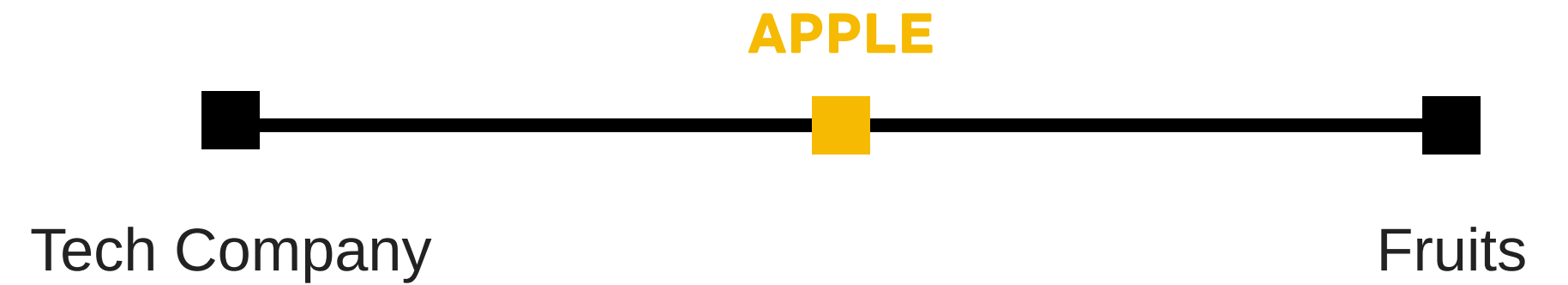
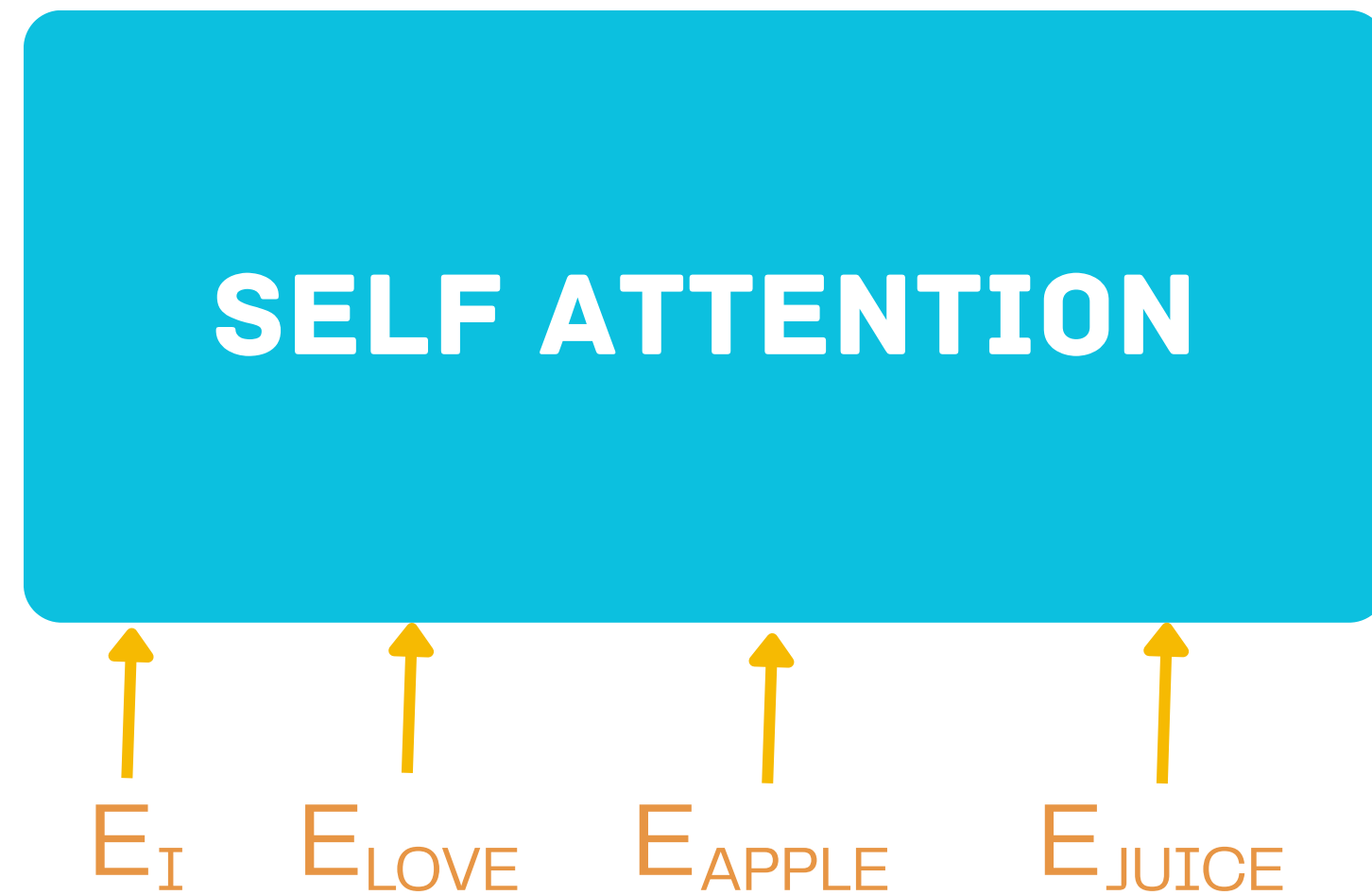
Self Attention - Brief Overview



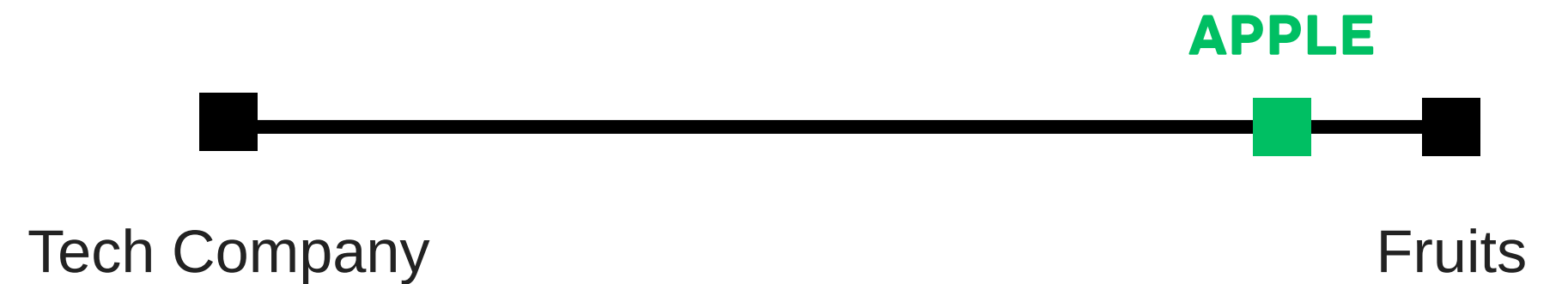
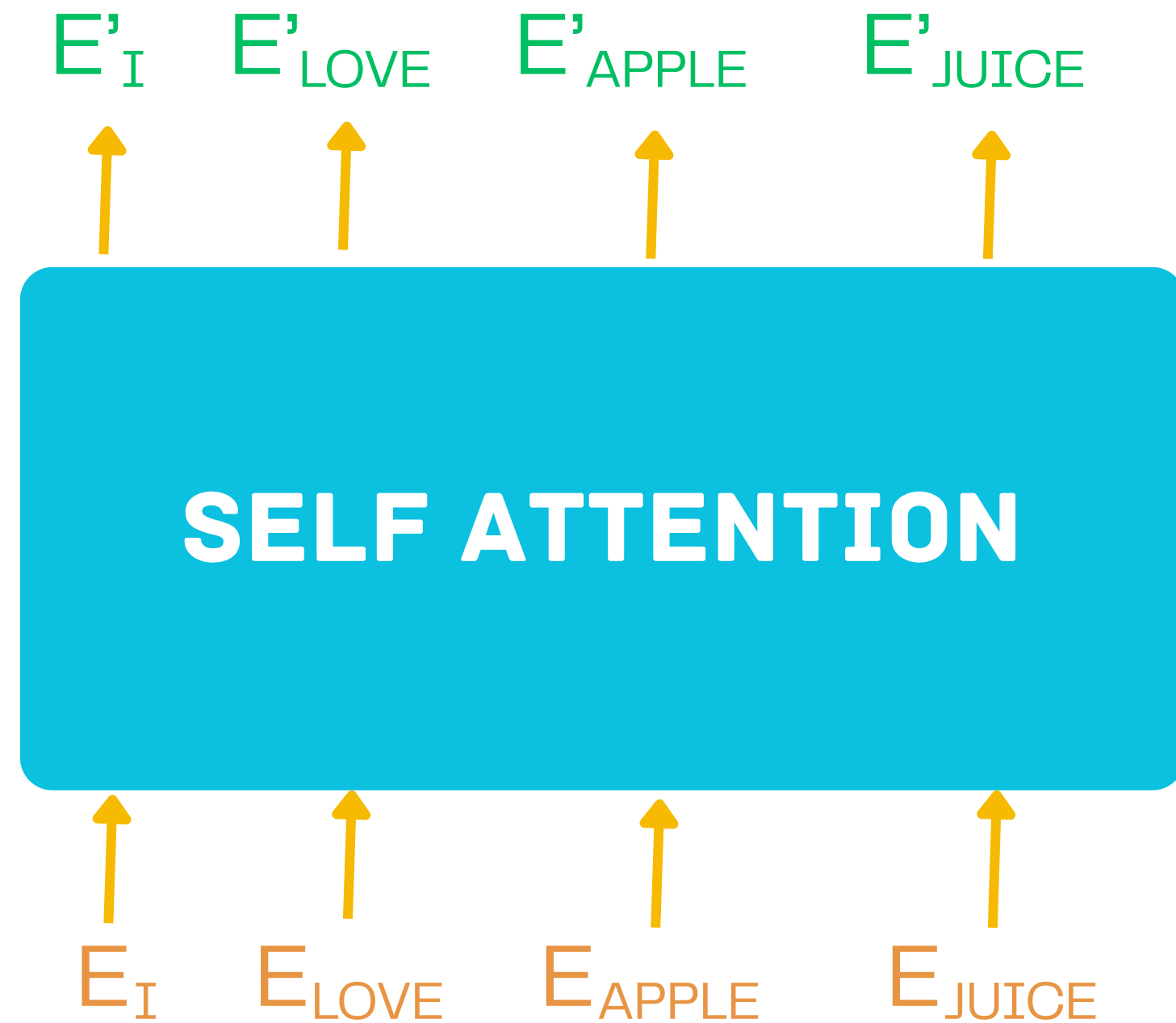
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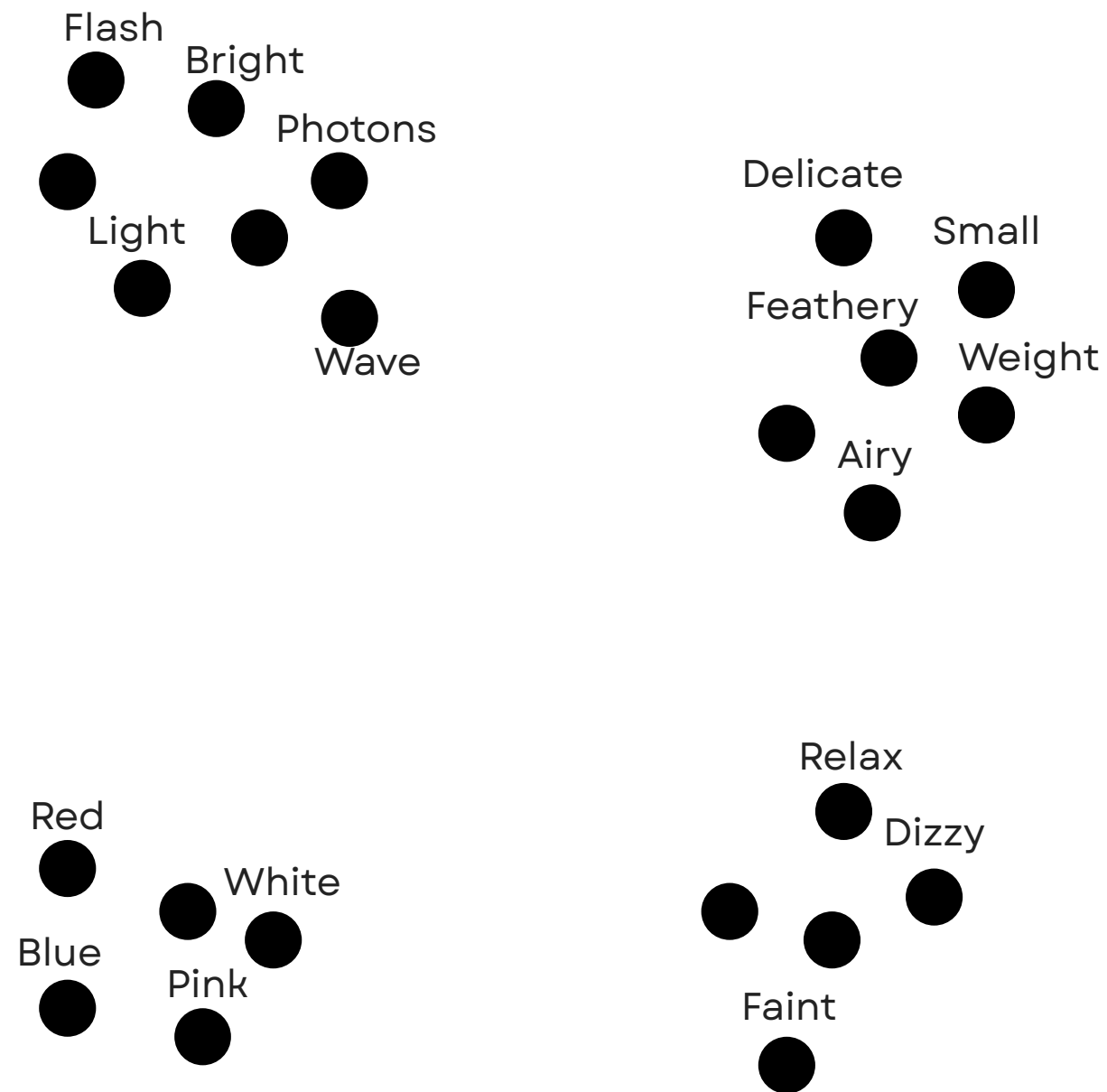


Self Attention - Brief Overview



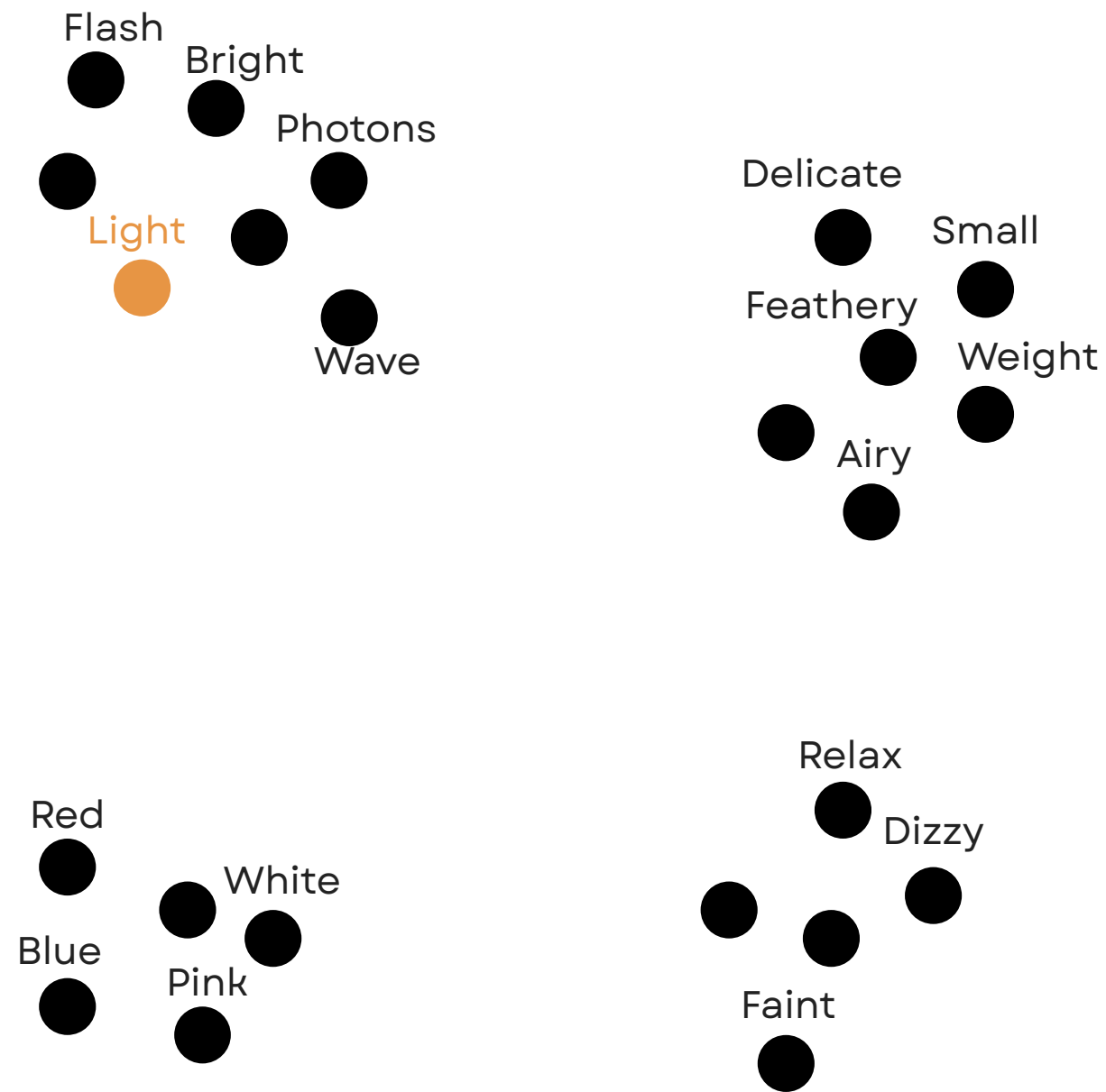
Self Attention - Brief Overview

Word Embeddings



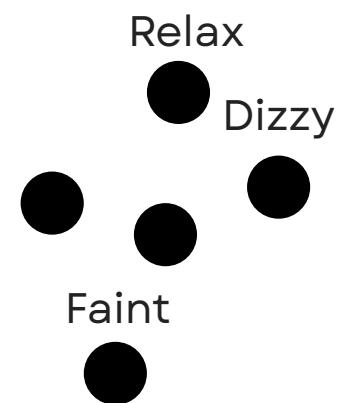
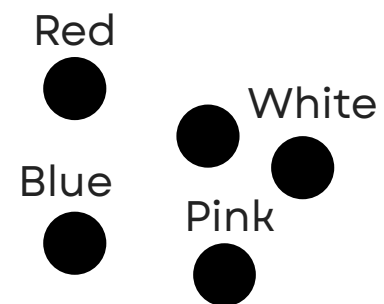
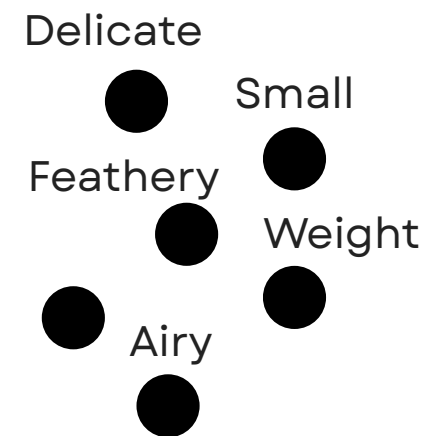
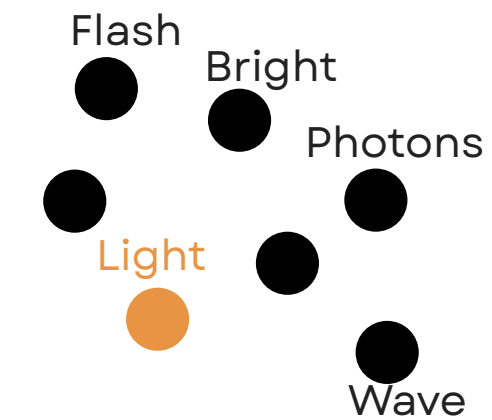
Self Attention - Brief Overview

Word Embeddings



Self Attention - Brief Overview

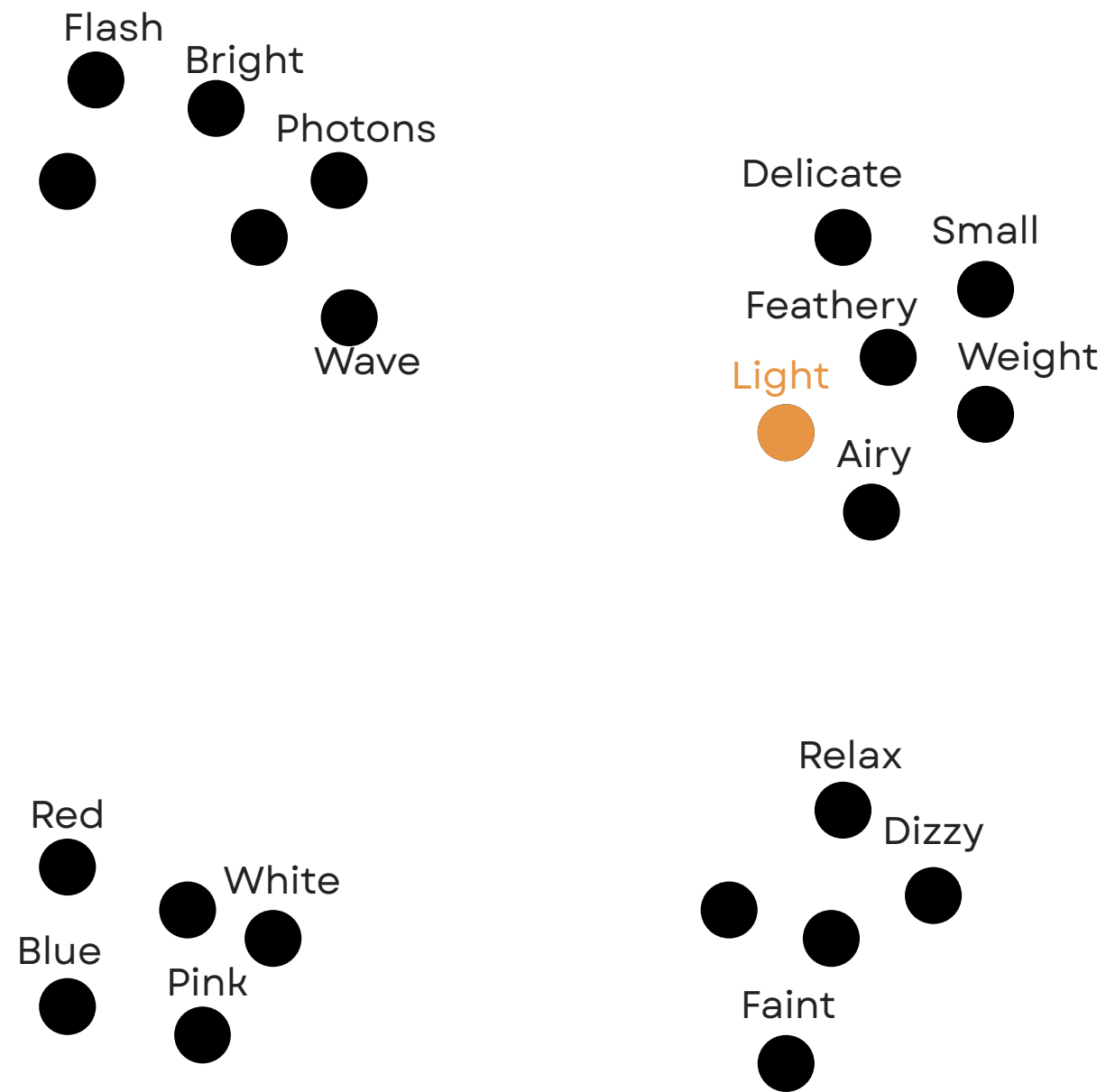
Word Embeddings




LIGHT WEIGHT

Self Attention - Brief Overview

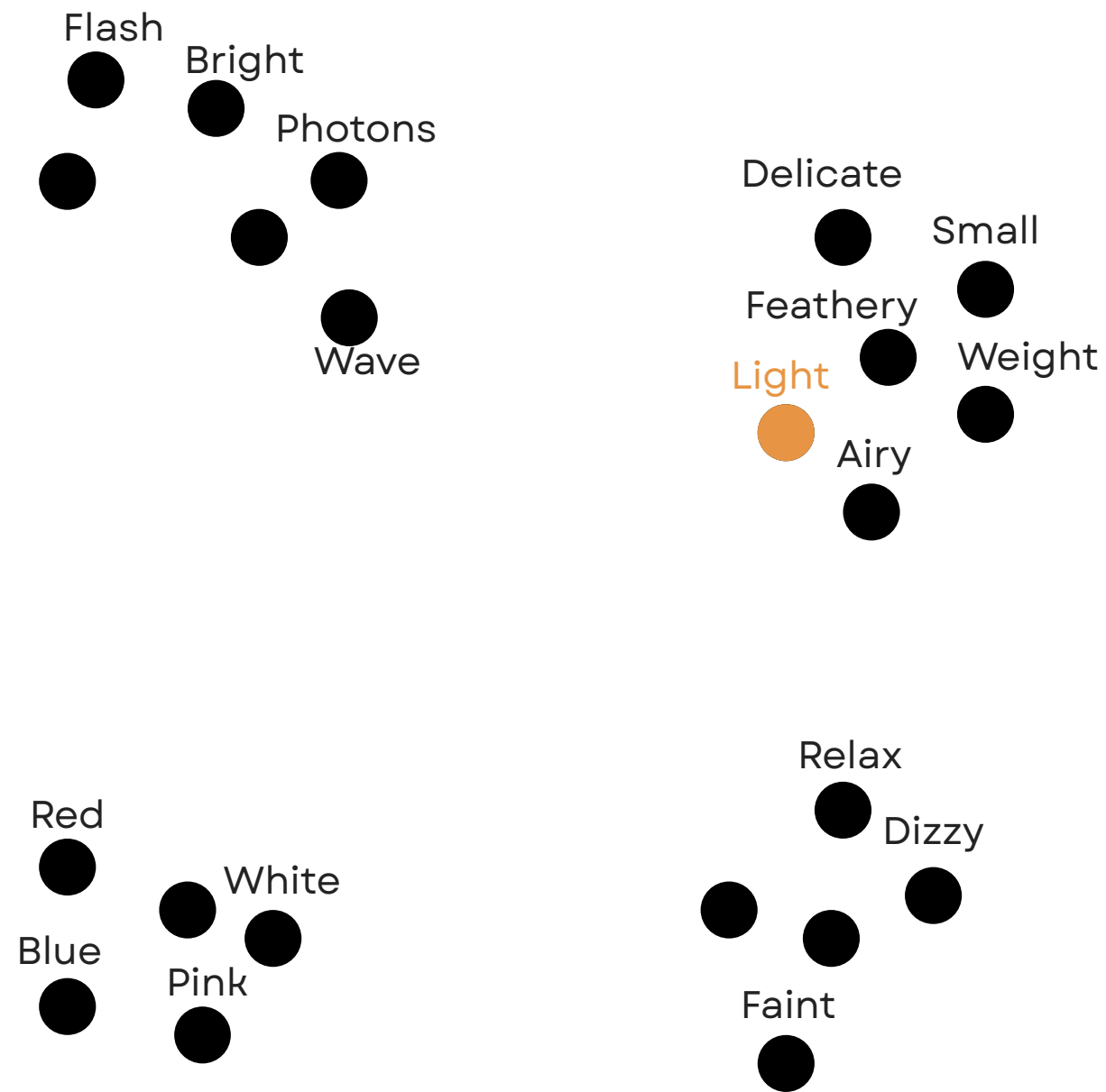
Word Embeddings




LIGHT WEIGHT

Self Attention - Brief Overview

Word Embeddings



LIGHT WEIGHT

LIGHT BLUE

Self Attention - Brief Overview

Word Embeddings

Flash
Bright
Photons
Wave

Delicate
Small
Feathery
Weight
Airy

Red
Blue
Light
White
Pink

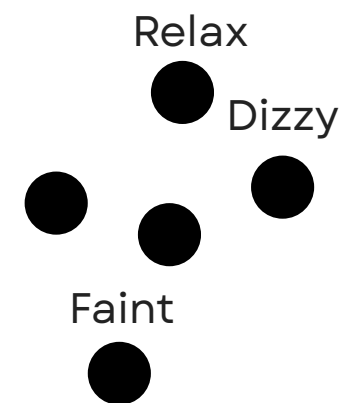
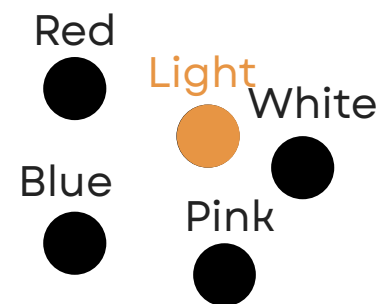
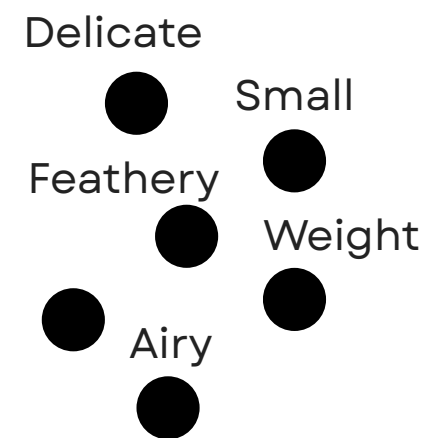
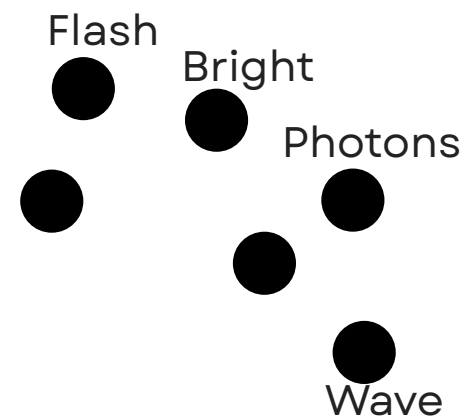
Relax
Dizzy
Faint


LIGHT WEIGHT

LIGHT BLUE

Self Attention - Brief Overview

Word Embeddings



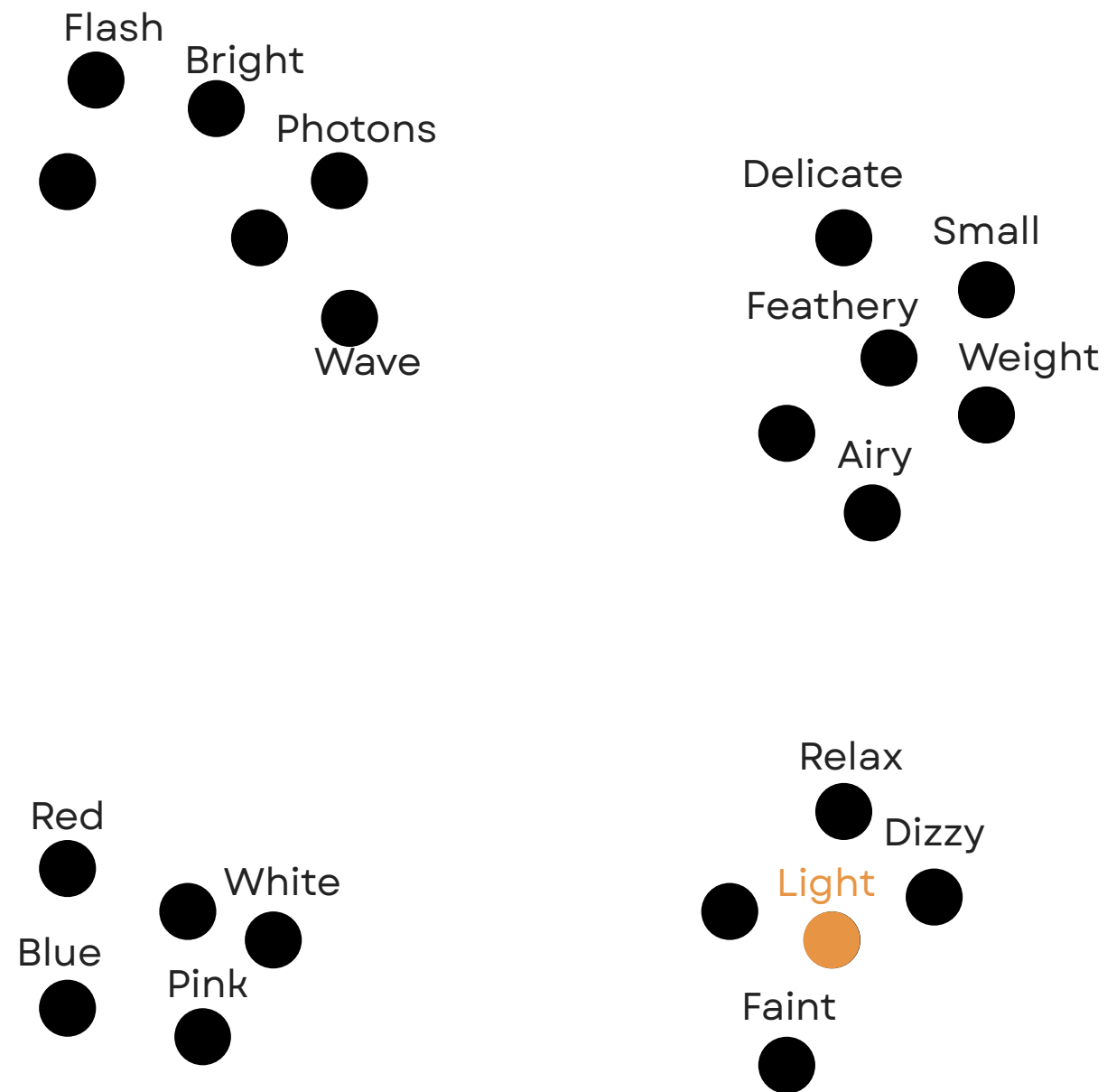

LIGHT WEIGHT


LIGHT BLUE


LIGHT HEADED

Self Attention - Brief Overview

Word Embeddings




LIGHT WEIGHT


LIGHT BLUE


LIGHT HEADED

Self Attention - Brief Overview

Generate an image of a leafy cloud



CLOUD

Self Attention - Brief Overview

Generate an image of a leafy cloud



CLOUD



LEAFY

Self Attention - Brief Overview

Generate an image of a leafy cloud



CLOUD

+



LEAFY

=



LEAFY CLOUD

Self Attention - Brief Overview

She saw a man with the telescope

Self Attention - Brief Overview

She saw a man with the telescope

1- She saw a man who was
holding a telescope



Self Attention - Brief Overview

She saw a man with the telescope

1- She saw a man who was holding a telescope



2- She used a telescope to see the man



Scale of Transformers

175 Billion Parameters

350GB

GPT-3

Scale of Transformers

175 Billion Parameters

350GB

GPT-3

1.8 Trillion Parameters

3.6 TB

GPT-4

Scale of Transformers

175 Billion Parameters

350GB

GPT-3

1.8 Trillion Parameters

3.6 TB

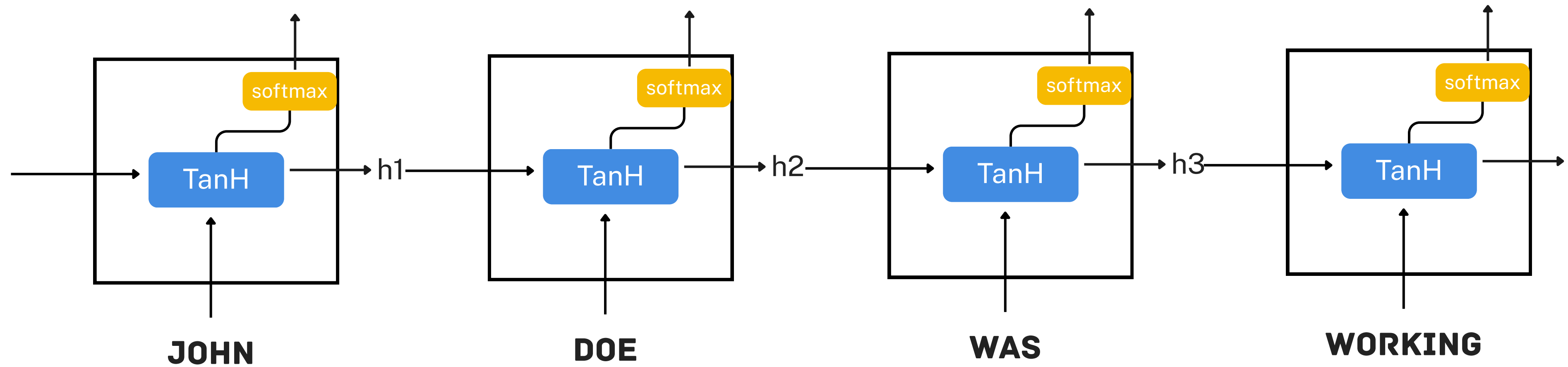
GPT-4

2-5 Trillion Parameters

4-10 TB

GPT-5

RNN



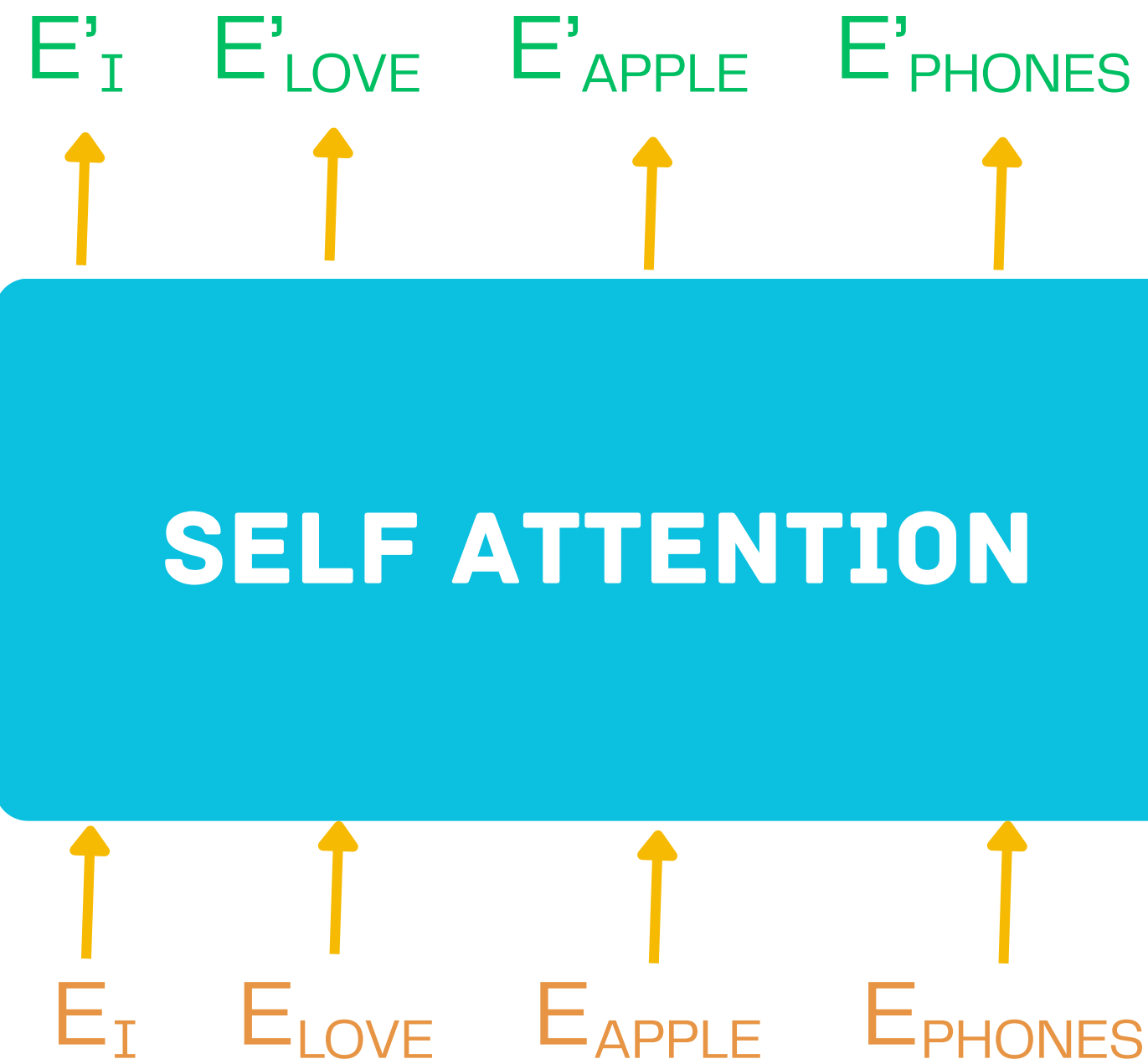
SELF ATTENTION

E_I

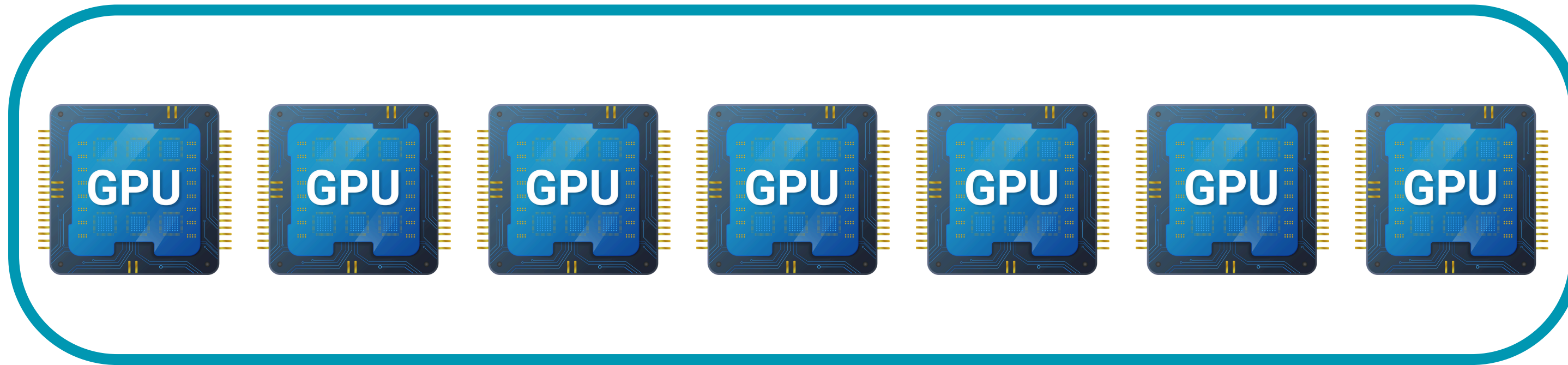
E_{LOVE}

E_{APPLE}

E_{PHONES}



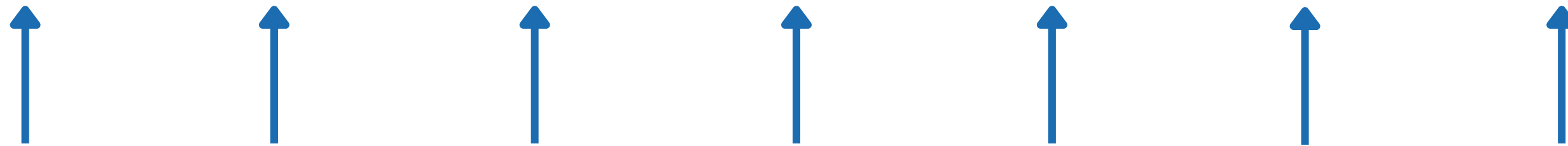
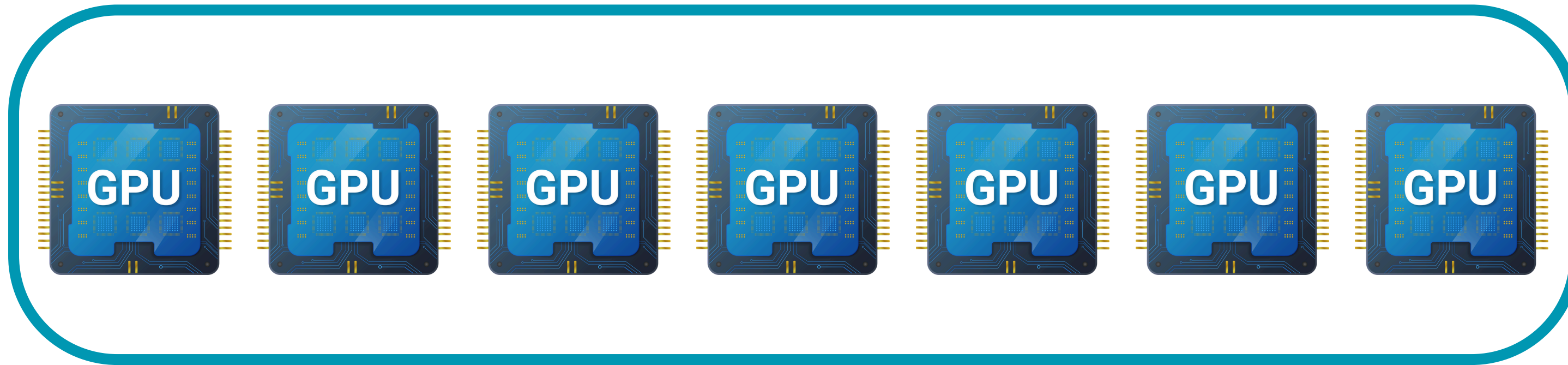
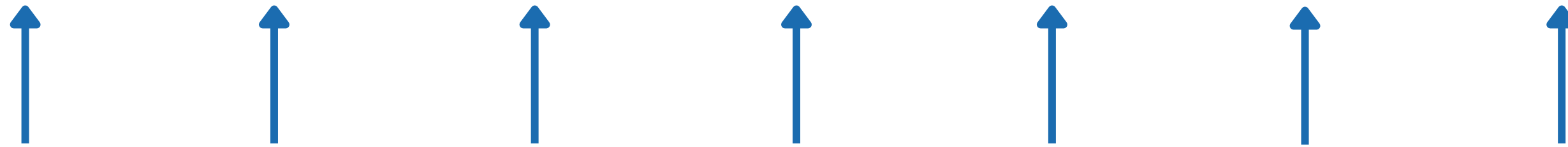
Parallelism in Transformers



When Jake first laid eyes on the Apple iPhone, he wasn't expecting much. It felt

Parallelism in Transformers

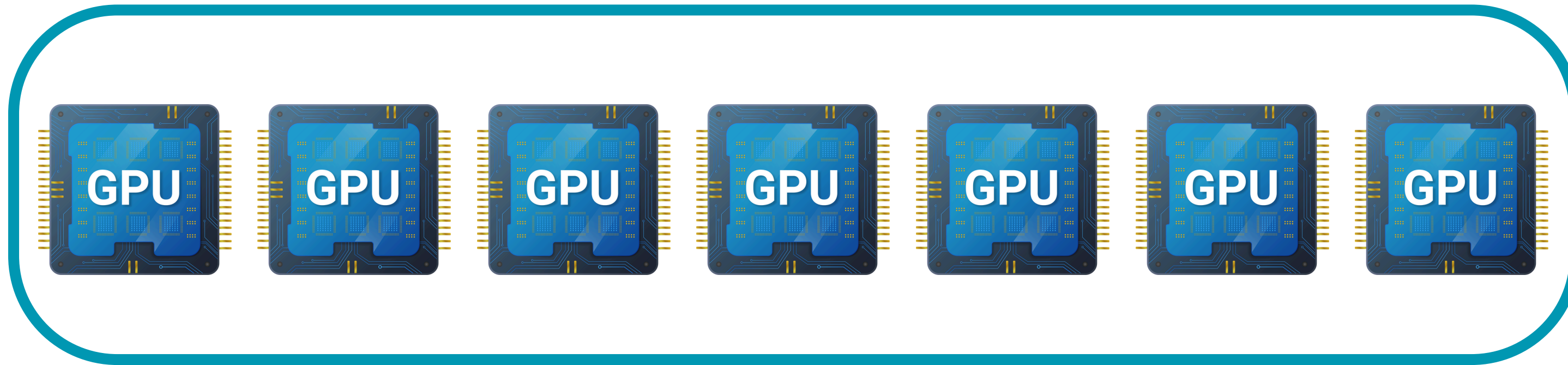
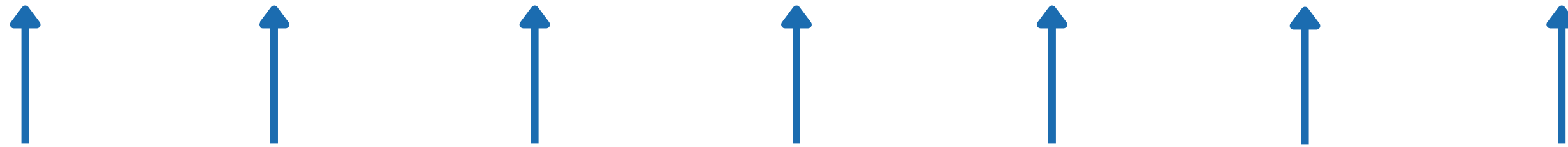
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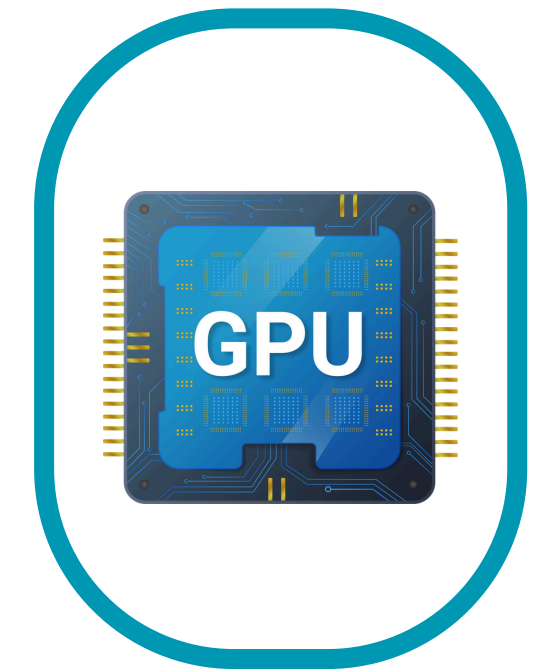
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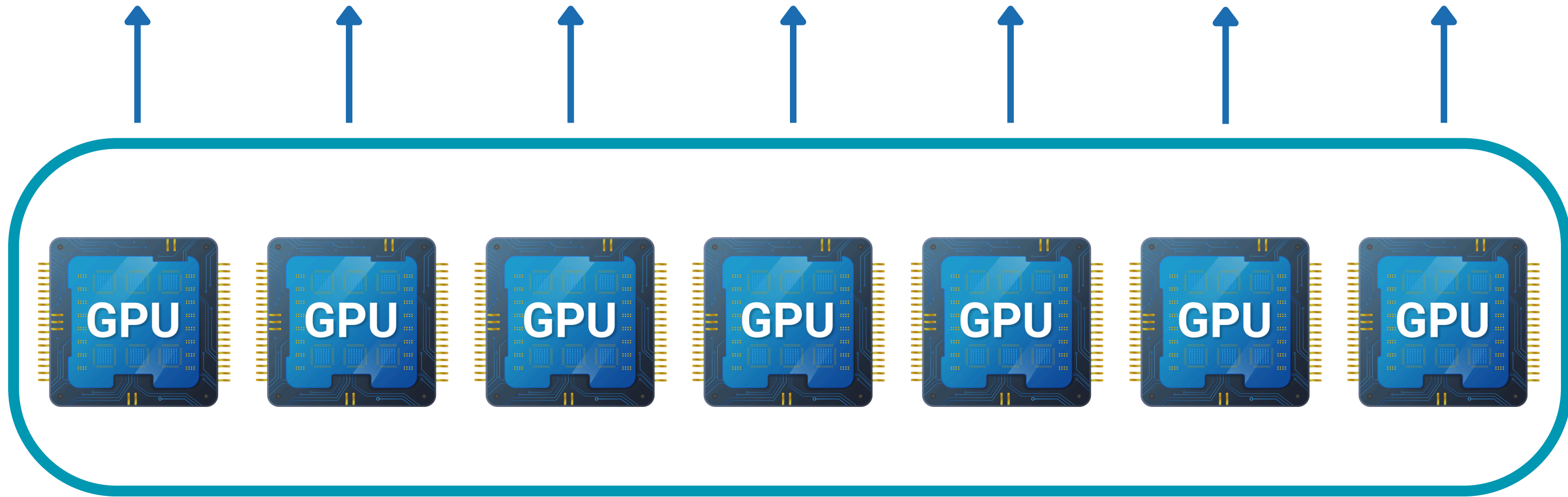
When



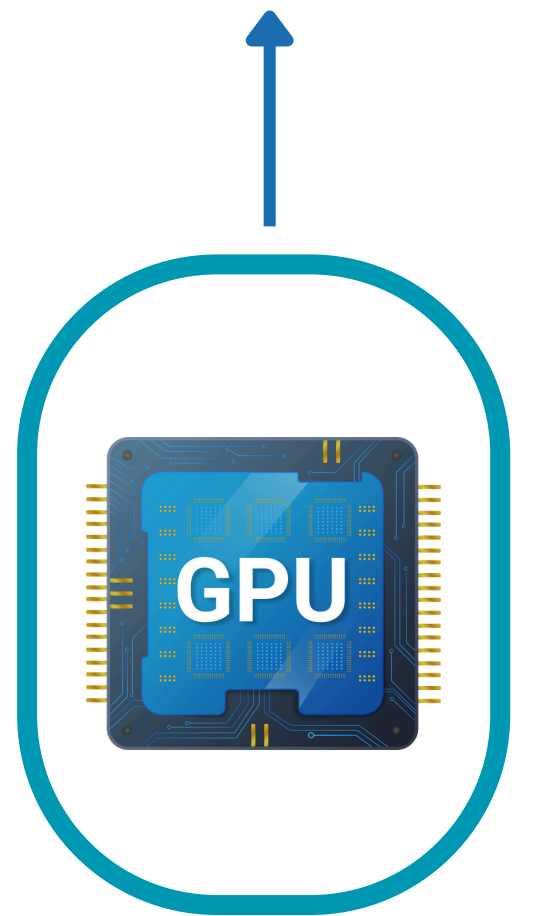
When

Parallelism in Transformers

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Jake



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Jake

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Apple

colorful

touchscreen

computer

phone

emails

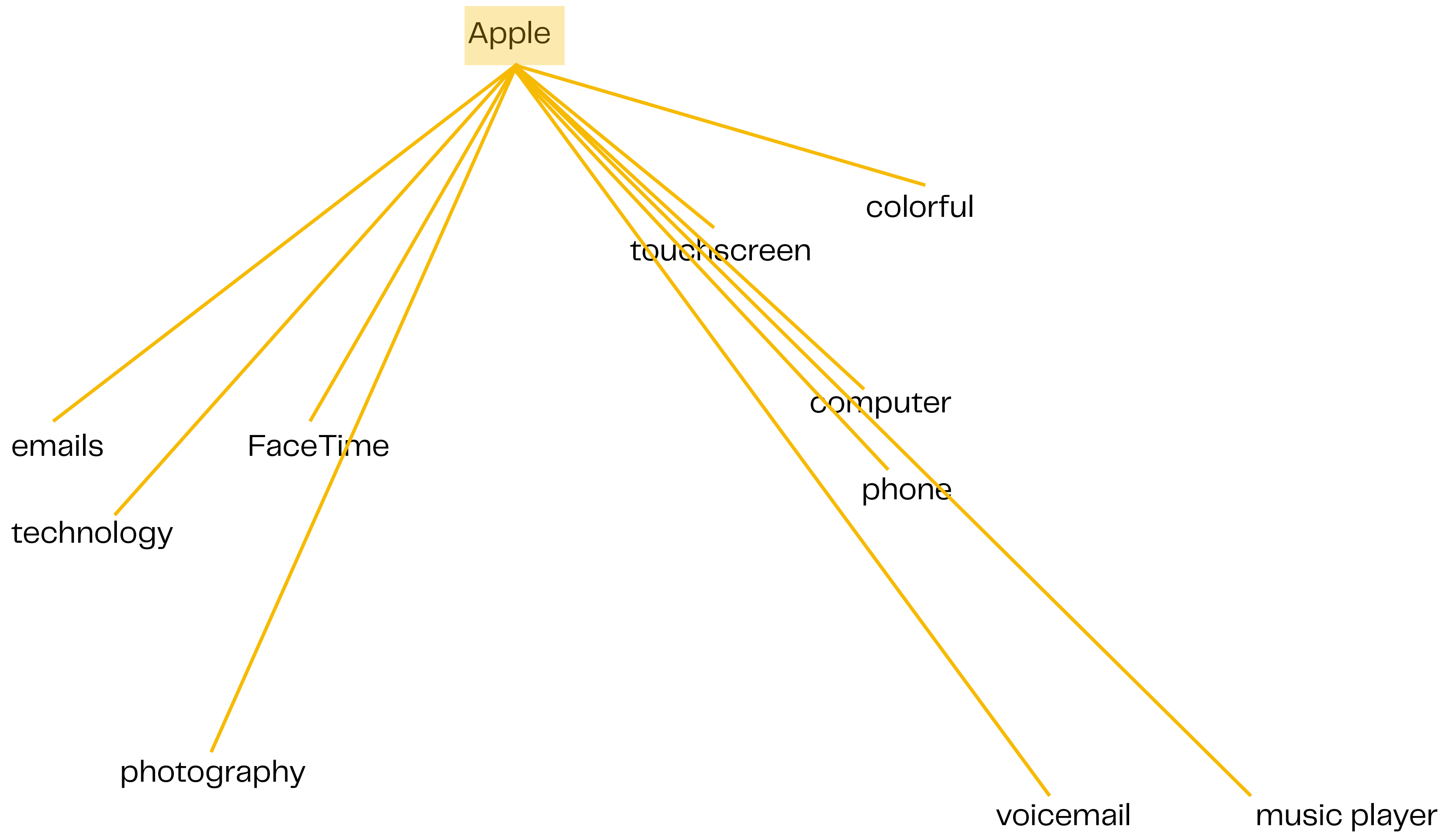
FaceTime

technology

photography

voicemail

music player



Transfer Learning

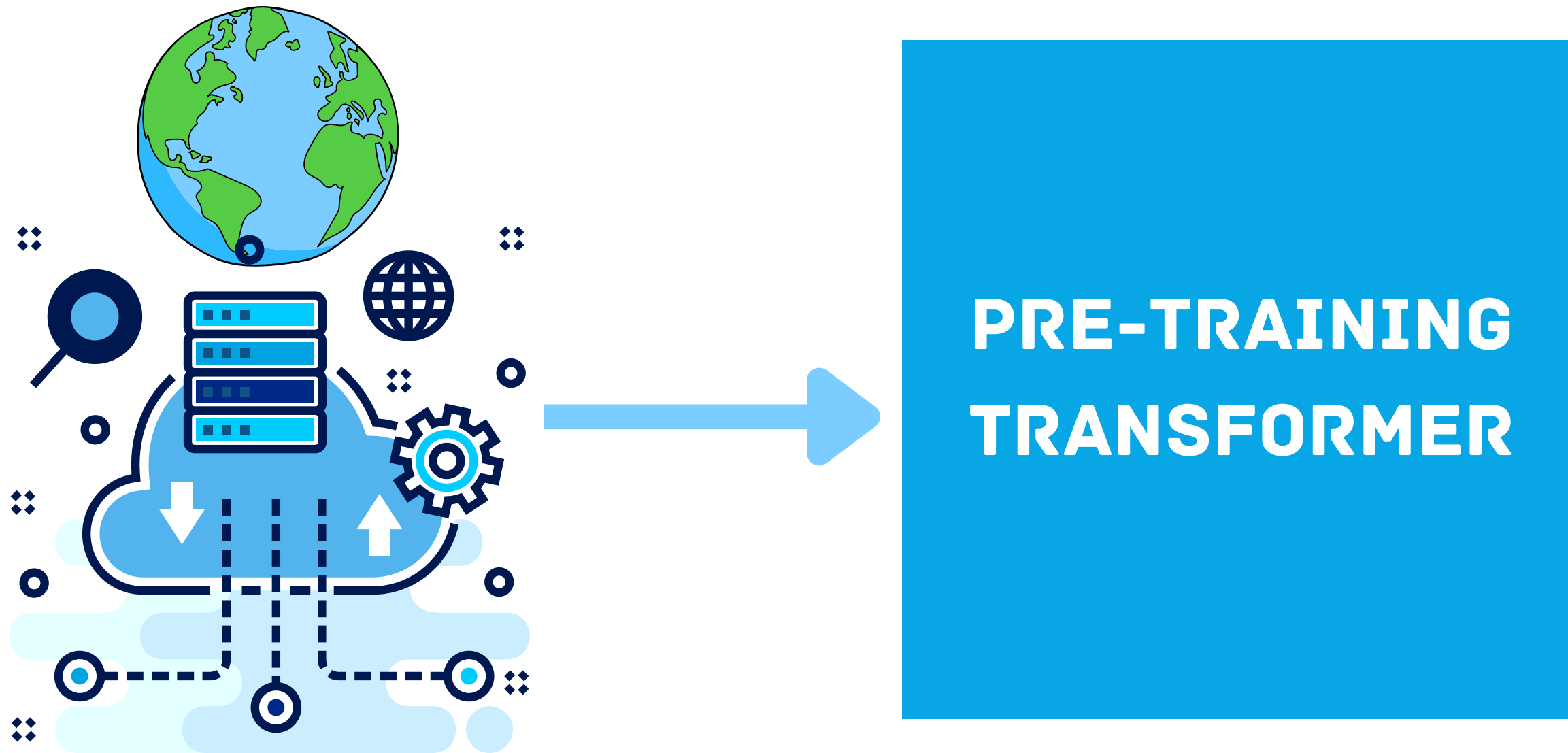


PRE-TRAIN

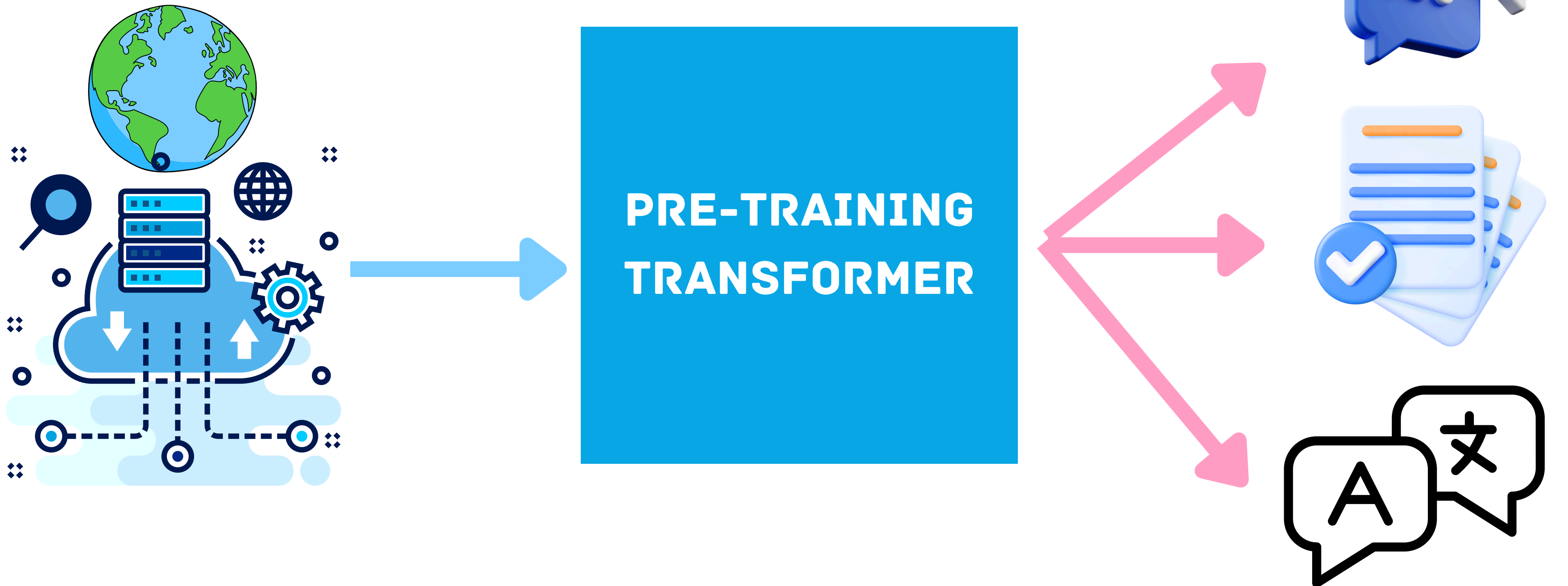


FINE-TUNE

Transfer Learning



Transfer Learning





The AI community building the future.

The platform where the machine learning community collaborates on models, datasets, and applications.

[Explore AI Apps](#)

or

[Browse 1M+ models](#)

Tasks Libraries Datasets Languages Licenses Other

Filter Tasks by name

Multimodal

- Text-to-Image
- Image-to-Text
- Text-to-Video
- Visual Question Answering
- Document Question Answering
- Graph Machine Learning

Computer Vision

- Depth Estimation
- Image Classification
- Object Detection
- Image Segmentation
- Image-to-Image
- Unconditional Image Generation
- Video Classification
- Zero-Shot Image Classification

Natural Language Processing

- Text Classification
- Token Classification
- Table Question Answering
- Question Answering
- Zero-Shot Classification
- Translation
- Summarization
- Conversational
- Text Generation
- Text2Text Generation
- Sentence Similarity

Audio

- Text-to-Speech
- Automatic Speech Recognition
- Audio-to-Audio
- Audio Classification
- Voice Activity Detection

Tabular

- Tabular Classification
- Tabular Regression

Reinforcement Learning

- Reinforcement Learning
- Robotics

Models 469,541 Filter by name

meta-llama/Llama-2-70b

Text Generation • Updated 4 days ago • 25.2k • 64

stabilityai/stable-diffusion-xl-base-0.9

Updated 6 days ago • 2.01k • 393

openchat/openchat

Text Generation • Updated 2 days ago • 1.3k • 136

lillyasviel/ControlNet-v1-1

Updated Apr 26 • 1.87k

cerspense/zeroscope_v2_XL

Updated 3 days ago • 2.66k • 334

meta-llama/Llama-2-13b

Text Generation • Updated 4 days ago • 328 • 64

tiiuae/falcon-40b-instruct

Text Generation • Updated 27 days ago • 288k • 899

WizardLM/WizardCoder-15B-V1.0

Text Generation • Updated 3 days ago • 12.5k • 332

CompVis/stable-diffusion-v1-4

Text-to-Image • Updated about 17 hours ago • 448k • 5.72k

stabilityai/stable-diffusion-2-1

Text-to-Image • Updated about 17 hours ago • 782k • 2.81k

Salesforce/xgen-7b-8k-inst

Text Generation • Updated 4 days ago • 6.18k • 57

Trending on 🤖 this week

📦 Models

zai-org/GLM-4.5V

Updated about 4 hours ago • ⬇️ 7.11k • ❤️ 514

openai/gpt-oss-20b

Updated 4 days ago • ⬇️ 3.42M • ❤️ 3.07k

google/gemma-3-270m

Updated 4 days ago • ⬇️ 2.92k • ❤️ 361

tencent/Hunyuan-GameCraft-1.0

Updated 4 days ago • ❤️ 341

openai/gpt-oss-120b

Updated 4 days ago • ⬇️ 788k • ❤️ 3.45k

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🧩 Spaces

DeepSite v2 🧪

❤️ 12k

Generate any application with DeepSeek

Qwen Image 🖼️

❤️ 633

Generate images from text prompts

Wan2.2 14B Fast 🎬

❤️ 192

generate a video from an image with a text prompt

GPT-OSS-120B on AMD MI300X 🖥️

❤️ 222

gpt-oss-120b model running on AMD MI300 infrastructure.

Ovis2.5 9B 📊

❤️ 102

High-accuracy vision & reasoning for complex tasks

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📁 Datasets

fka/awesome-chatgpt-prompts

Updated Jan 6 • ⬇️ 34.8k • ❤️ 8.73k

nvidia/Llama-Nemotron-VLM-Dataset-v1

Updated 13 days ago • ⬇️ 1.23k • ❤️ 84

allenai/WildChat-4.8M

Updated 7 days ago • ⬇️ 1.19k • ❤️ 69

nvidia/Granary

Updated 4 days ago • ⬇️ 149 • ❤️ 53

miromind-ai/MiroVerse-v0.1

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Multi-Modality

TEXT

Multi-Modality

TEXT



Image

Multi-Modality

TEXT



Image



Audio

Multi-Modality

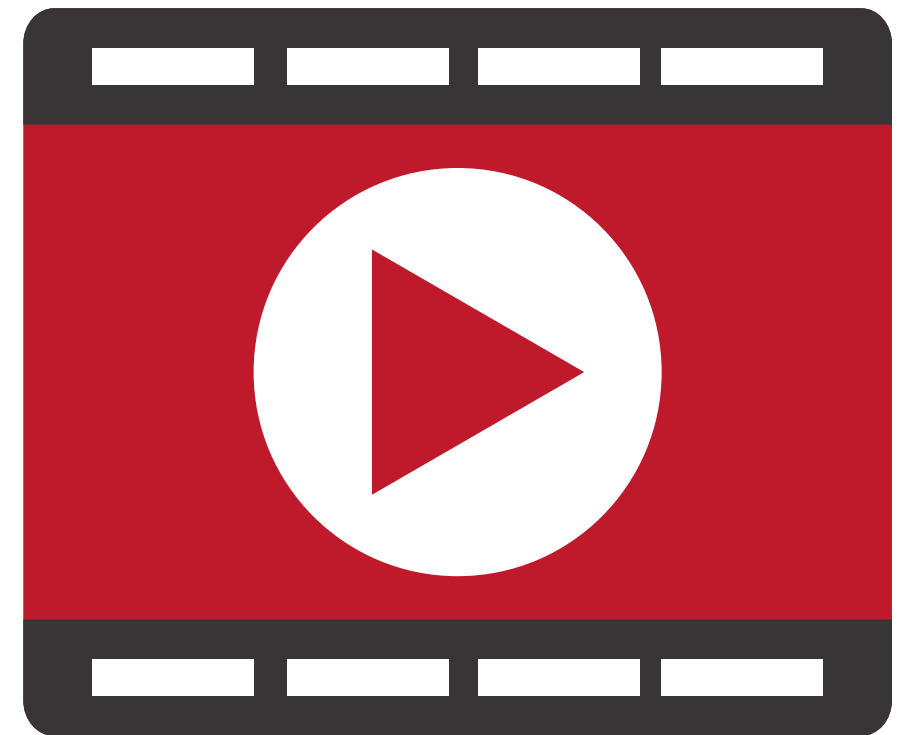
TEXT



Image



Audio



Video

Multi-Modality

Generate an image
of a man with the
telescope



TRANSFORMER

Multi-Modality

TRANSFORMER



Multi-Modality



TRANSFORMER

Multi-Modality



TRANSFORMER

A woman holding
an arrow

Multi-Modality



Find me a similar
chair online

TRANSFORMER

Multi-Modality

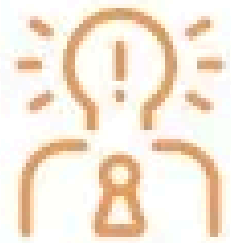


TRANSFORMER

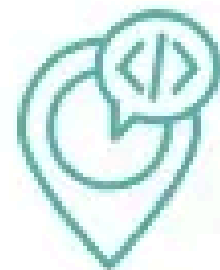
sure, here are some
results

Core Concepts Of Transformers

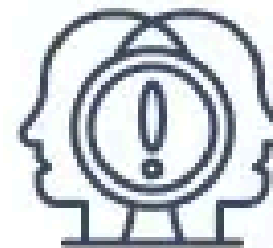
Self-Attention



Positional Encoding



Multi-Head Attention



Encoder-Decoder

